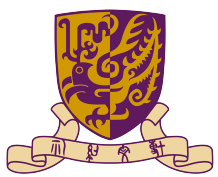
Globally Converging Cutting-Plane Algorithms for Multistage Stochastic Mixed-Integer Programming with Applications in Power Systems

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School of Data Science
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May 9, 2025



Outline

1 Introduction

2 Lift-and-Cut Algorithm

3 Improving Benders Cuts via Disjunctive Programming

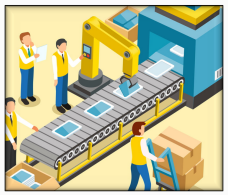
4 Two-stage Model for Mitigating Wildfire Risk

5 Summary

Optimization Under Uncertainty

- Real-world decision problems: dynamic and affected by uncertainty

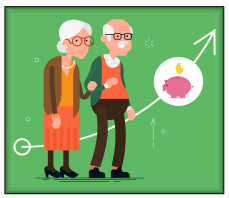

Finance


Production


Transportation


Logistics


Energy


Pension Plans

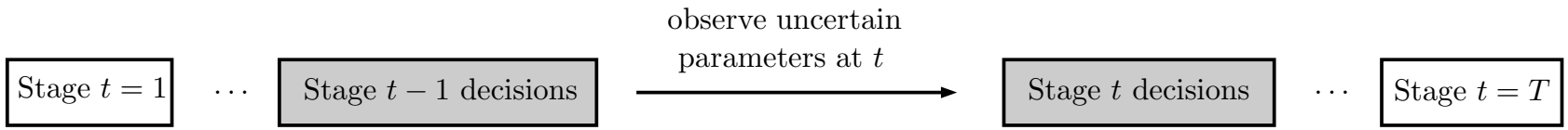
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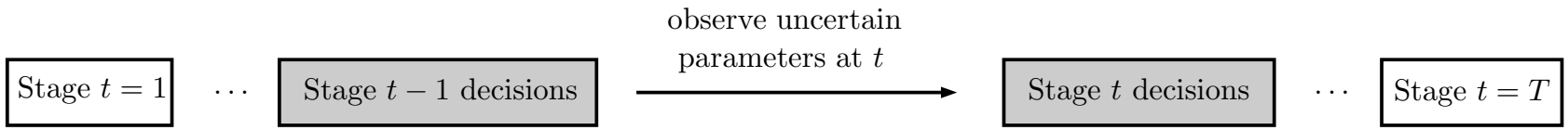


Domain	Application	Reference(s)	Uncertainty Example
Energy	Unit Commitment	Bienstock et al. [2024], Zou [2017]	Load demand, renewable generation
Transportation	Air Traffic Management	Wang and Jacquillat [2020]	Weather, air traffic demand
Scheduling	Healthcare Surgery Scheduling	Gul et al. [2015], Guo et al. [2021], Kim and Mehrotra [2015]	Patient arrivals, surgery duration

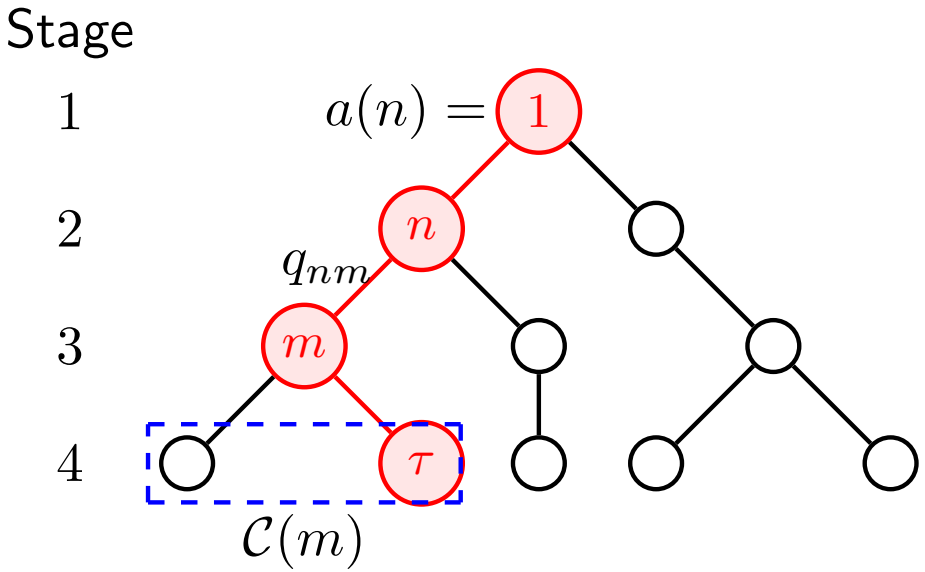
Multistage Stochastic Programming



Multistage Stochastic Programming



Describe the stochastic process $(\tilde{\boldsymbol{\xi}}_1, \ldots, \tilde{\boldsymbol{\xi}}_T)$ via a scenario tree $\mathcal{N}$:



Node 1: root node;
$a(n)$: parent node of n;
q_{nm}: transition probability;
$\mathcal{C}(n)$: set of children nodes of n;
$\mathcal{P}^\omega = \{1, n, m, \tau\}$: scenario path.

Dynamic Programming Formulation

The recursive formulation with expected cost-to-go functions:

$$V^* = \min_{x_1, y_1} \quad \overbrace{c_1^\top x_1 + g_1^\top y_1}^{\text{current cost}} + \overbrace{\sum_{m \in \mathcal{C}(1)} q_{1m} Q_m(x_1)}^{\text{future cost}} \tag{P}$$

$$\text{s.t.} \quad W_1 x_1 + B_1 y_1 = d_1$$

$$x_1 \in X_1, y_1 \in Y_1.$$

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The nodal subproblem at node $n \in \mathcal{N}$ is defined as:

$$Q_n(x_{a(n)}) := \min_{x_n, y_n} \quad c_n^\top x_n + g_n^\top y_n + \sum_{m \in \mathcal{C}(n)} q_{nm} Q_m(x_n)$$

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Properties of Q_n (rational, compact MILP sets X_n, Y_n):

- $\nexists$ Int. Var.: $Q_n(\cdot)$ is piecewise linear and convex.

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- $\nexists$ Int. Var.: $Q_n(\cdot)$ is piecewise linear and convex.
- $\exists$ Int. Var.: $Q_n(\cdot)$ is non-convex and lower semi-continuous (l.s.c.).

Literature Review

- **Stochastic Dual Dynamic Programming (SDDP) [Pereira and Pinto, 1991]**
  - Assumption: No integer variables;
  - Generate Benders cuts to approximate value function Q_m;

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- **SDDP+Lipschitz Cuts [Ahmed et al., 2020]**
  - Assumption: Lipschitz Continuity of Q_m;
  - Approximate the Lip. value functions by Lipschitz cuts;
  - $\rightsquigarrow$ Complete continuous recourse $\not\Rightarrow$ General mixed-integer variables.

Value Function Approximation in SDDP

- ϕ_m is an under-approximation of Q_m;
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$$\phi_m(x_n) := \min \left\{ \theta_m \;\middle|\; \theta_m \geq v_m^\ell + {\pi_m^\ell}^\top x_n, \quad \forall \ell = 0, \ldots, L_m \right\}$$

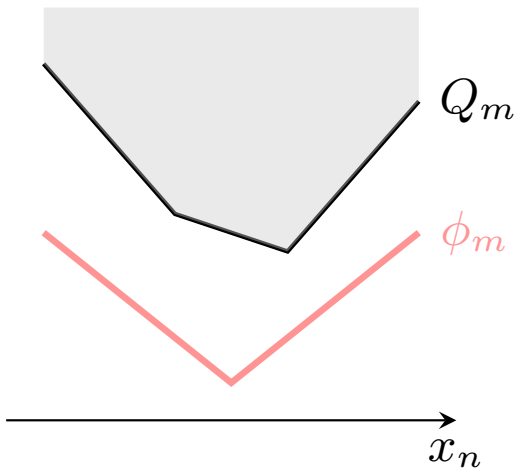


Figure: Q_m is convex.

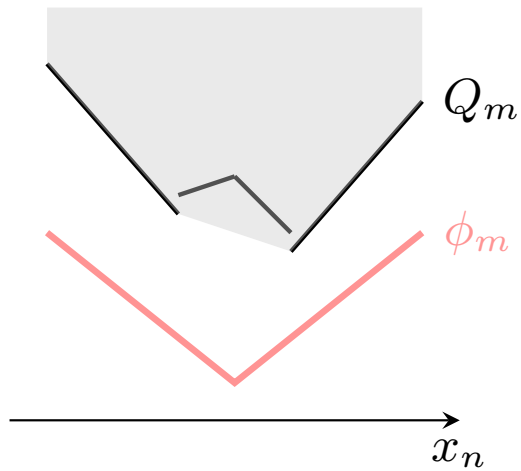


Figure: Q_m is l.s.c.

Approximate Nodal Subproblem

Approximate $\{Q_m\}_{m\in\mathcal{C}(n)}$ by cut approximators $\Phi_n := \{\phi_m\}_{m\in\mathcal{C}(n)}$:

$$\underline{Q}_n(x_{a(n)};\Phi_n) :=$$

$$\begin{aligned} \min \quad & c_n^\top x_n + g_n^\top y_n + \sum_{m\in\mathcal{C}(n)} q_{nm} \cancel{Q_m(x_n)} \phi_m(x_n) && (P_n) \\ \text{s.t.} \quad & W_n x_n + B_n y_n = T_n \cancel{x_{a(n)}}\, z_n + d_n \\ & x_n \in X_n, y_n \in Y_n \\ & z_n = x_{a(n)} && \text{(NA)} \end{aligned}$$

where z_n is the copy variable with the non-anticipativity constraint.

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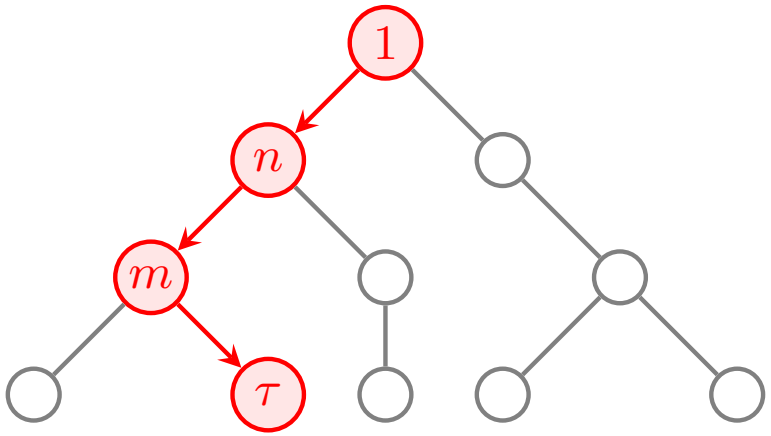
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- $\underline{Q}_n(\cdot;\Phi_n)$ under-approximates the true value function Q_n;
- Constraint (NA) is used to generate cuts that improve the approximation;
- Solve (P_n) in SDDP algorithm.

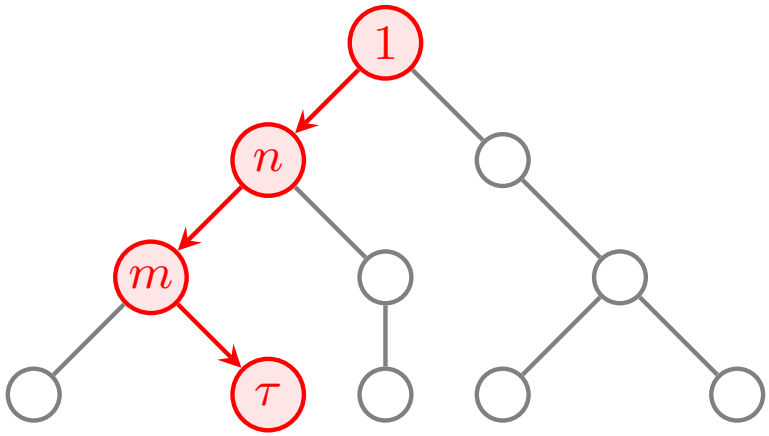
SDDP – Forward Pass

1. Sample a scenario path $\mathcal{P}^\omega = \{1, n, m, \tau\}$;



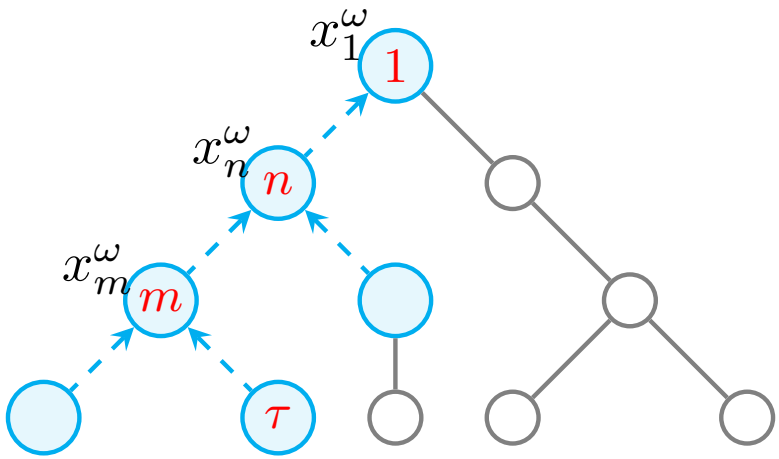
SDDP – Forward Pass

1. Sample a scenario path $\mathcal{P}^\omega = \{1, n, m, \tau\}$;



2. **For** each node $n \in \mathcal{P}^\omega$:
   - Solve the approximate nodal subproblem (P_n);
   - Collect the decisions (x_n^ω, y_n^ω).

SDDP – Backward Pass



- **For** each node $n \in \mathcal{P}^\omega$:
  - **For** each child node $m \in \mathcal{C}(n)$:

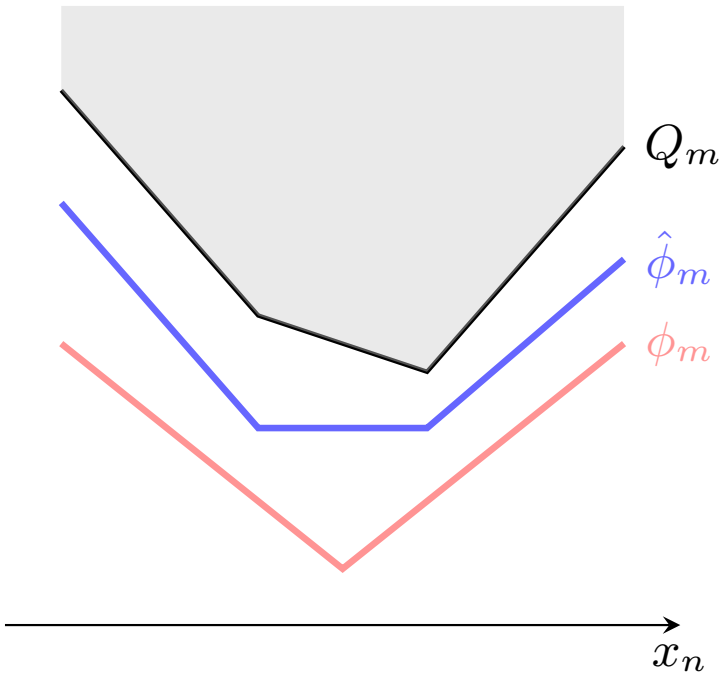
    Generate a cut of the form

    $$\theta_m \geq v_m + \pi_m^\top x_n$$

    to refine the approximator ϕ_m of the value function Q_m.

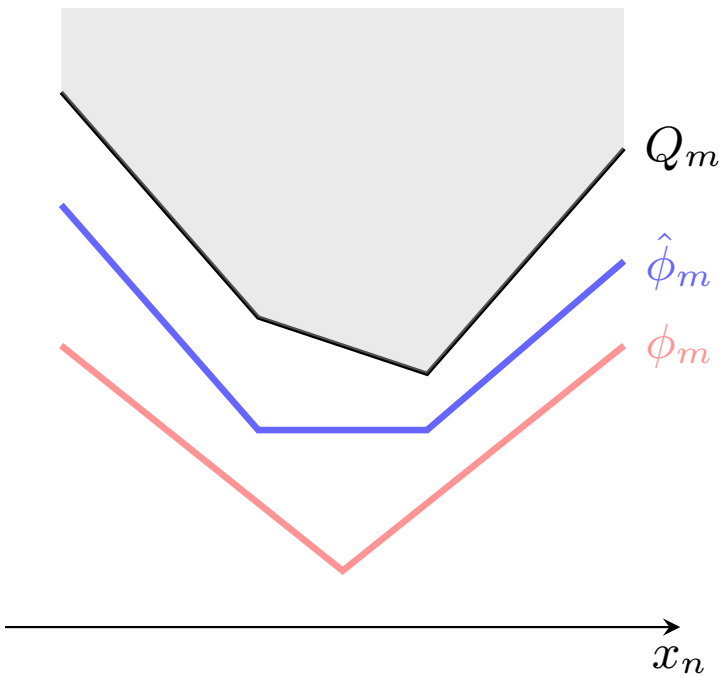
Refinement of Cut Approximator ϕ_m

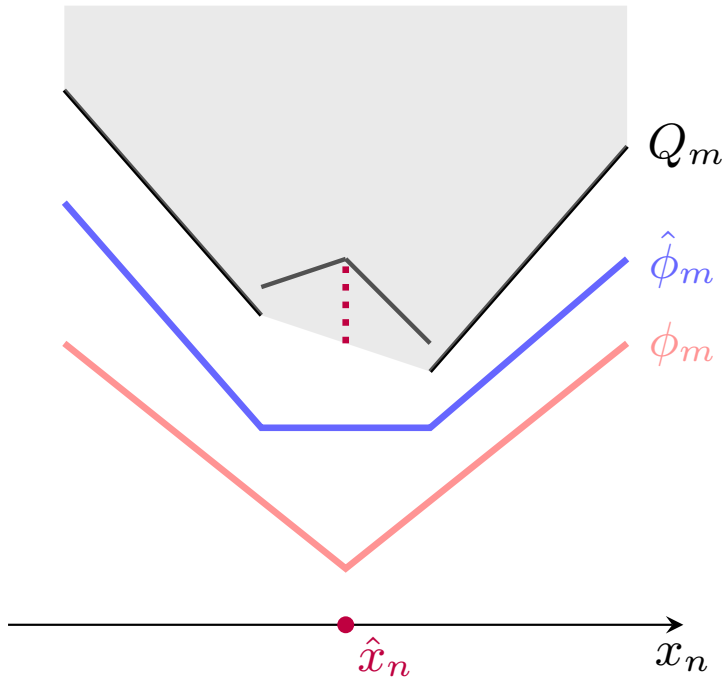
- ϕ_m is improved by incorporating cuts $\rightsquigarrow$ $\hat{\phi}_m$;



Refinement of Cut Approximator ϕ_m

- ϕ_m is improved by incorporating cuts $\rightsquigarrow$ $\hat{\phi}_m$;





- If Q_m is l.s.c., there always exists gap between ϕ_m and Q_m.

Definition

Let f be an l.s.c. function. The *epigraph* of f over $\mathcal{Z}$ is

$$\mathrm{epi}_{\mathcal{Z}}(f) = \{(x, \eta) \mid \eta \geq f(x)\ \forall x \in \mathcal{Z}\}.$$

The *convex envelope* of f over $\mathcal{Z}$ is

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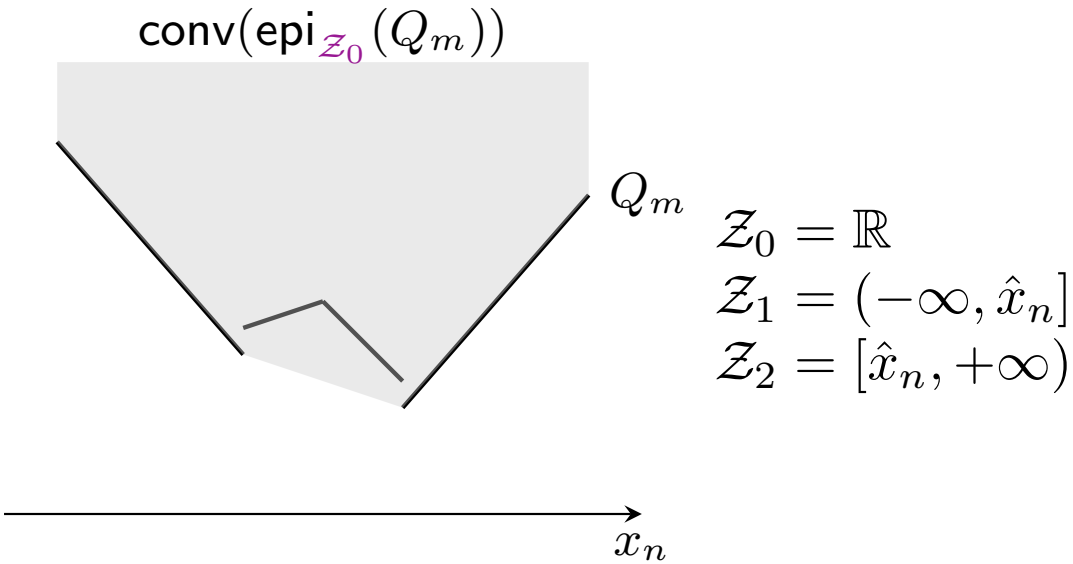
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$$\mathcal{Z}_0 = \mathbb{R}$$
$$\mathcal{Z}_1 = (-\infty, \hat{x}_n]$$
$$\mathcal{Z}_2 = [\hat{x}_n, +\infty)$$

Figure: Convex envelope over $\mathcal{Z}_0$.

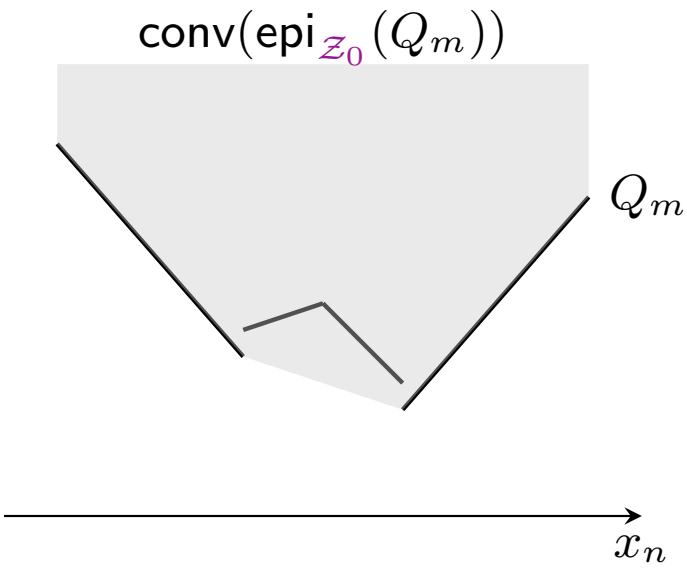
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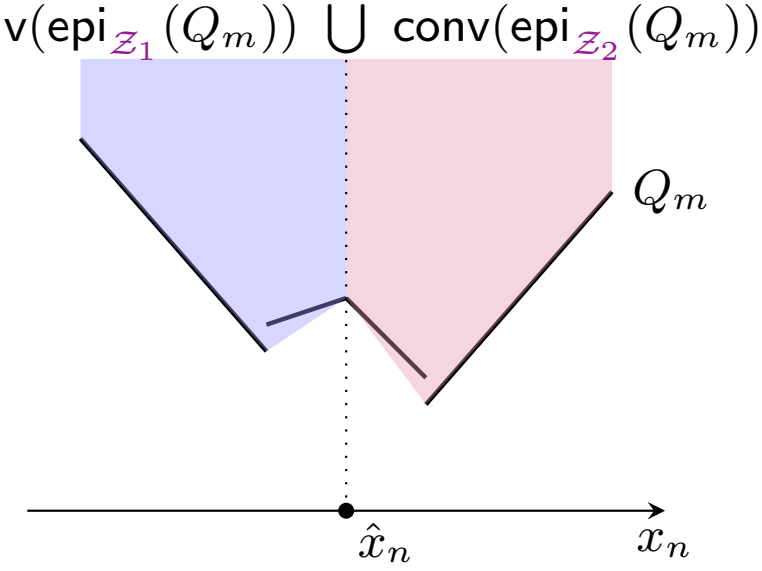


Figure: Convex envelope over $\mathcal{Z}_0$.

Figure: **Piecewise** convex envelope.

Lagrangian Cuts & Properties

Lagrangian-Cut Generation Program

Goal: Refine ϕ_m by adding new cuts.

Given: State $\hat{x}_n$ at parent node $n = a(m)$.

The Lagrangian dual problem is:

$$\max_{\pi_m} \quad \mathcal{L}_m\left(\pi_m; \hat{x}_n, \Phi_m, \mathcal{Z}_n\right), \tag{D_m}$$

where $\mathcal{L}_m(\pi_m; \hat{x}_n, \Phi_m, \mathcal{Z}_n) :=$

$$\begin{aligned} \min \quad & c_m^\top x_m + g_m^\top y_m + \sum_{\tau \in \mathcal{C}(m)} q_{m\tau}\phi_\tau(x_m) + \pi_m^\top(\hat{x}_n - z_m) \\ \text{s.t.} \quad & W_m x_m + B_m y_m = T_m z_m + d_m \\ & x_m \in X_m, y_m \in Y_m \\ & z_m \in \mathcal{Z}_n \end{aligned}$$

Note: Restrict z_m into a predefined region $\mathcal{Z}_n$.

Lagrangian Cuts & Properties

Given: An optimal dual solution π_m^* to (D_m).

Generate a Lagrangian cut:

$$\theta_m \geq v_m^* + {\pi_m^*}^\top x_n,$$

where $v_m^* := \mathcal{L}_m(\pi_m^*; \hat{x}_n, \Phi_m, \mathcal{Z}_n) - {\pi_m^*}^\top \hat{x}_n$.

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The Lagrangian cut $\theta_m \geq v_m^* + {\pi_m^*}^\top x_n$ is:

1. valid over $\mathcal{Z}_n$: $\forall\, x_n \in \mathcal{Z}_n$,

$$Q_m(x_n) \geq \underline{Q}_m(x_n; \Phi_m) \geq v_m^* + {\pi_m^*}^\top x_n;$$

2. tight at $\hat{x}_n$ for the convex envelope of $\underline{Q}_m(\cdot; \Phi_m)$ over $\mathcal{Z}_n$:

$$v_m^* + {\pi_m^*}^\top \hat{x}_n = \mathrm{co}_{\mathcal{Z}_n}\left(\underline{Q}_m(\cdot; \Phi_m)\right)(\hat{x}_n).$$

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From Lagrangian Cuts to Global Optimality

Motivation:

- Lagrangian cuts recover the convex envelope of $\underline{Q}_m$ over $\mathcal{Z}_n$;
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Key Insight:

- Partition domain into smaller subregions:
  - Use binary indicator variables to indicate the selected regions;

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Key Insight:

- Partition domain into smaller subregions:
  - Use binary indicator variables to indicate the selected regions;
- Generate Lagrangian cuts in the lifted space.

Partition State Space

Step 1: Partition the State Space

- Each state variable $x_{n,j}$ lies in an interval $[\underline{x}_{n,j}, \overline{x}_{n,j}]$;
- We divide this interval into smaller subintervals:

$$\mathcal{P}_{n,j} := \left\{ [\underline{x}^{\sigma}_{n,j}, \overline{x}^{\sigma}_{n,j}] \ : \ \sigma \in \mathcal{S}_{n,j} \right\}.$$

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Step 2: Add Binary Indicators

- For each $\sigma \in \mathcal{S}_{n,j}$, introduce a binary variable $\lambda^{\sigma}_{n,j} \in \{0, 1\}$;
- $\lambda^{\sigma}_{n,j} = 1$ means that $x_{n,j}$ lies in interval $[\underline{x}^{\sigma}_{n,j}, \overline{x}^{\sigma}_{n,j}]$.

Partition State Space

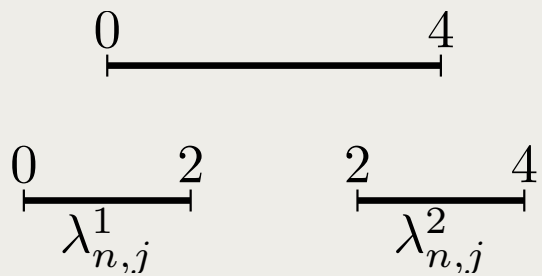
Step 1: Partition the State Space

- Each state variable $x_{n,j}$ lies in an interval $[\underline{x}_{n,j}, \overline{x}_{n,j}]$;
- We divide this interval into smaller subintervals:

$$\mathcal{P}_{n,j} := \left\{ [\underline{x}^{\sigma}_{n,j}, \overline{x}^{\sigma}_{n,j}] \ : \ \sigma \in \mathcal{S}_{n,j} \right\}.$$

Step 2: Add Binary Indicators

- For each $\sigma \in \mathcal{S}_{n,j}$, introduce a binary variable $\lambda^{\sigma}_{n,j} \in \{0, 1\}$;
- $\lambda^{\sigma}_{n,j} = 1$ means that $x_{n,j}$ lies in interval $[\underline{x}^{\sigma}_{n,j}, \overline{x}^{\sigma}_{n,j}]$.



Let $x_{n,j}$ be continuous over $[0, 4]$.

- $\mathcal{P}_{n,j} = \{[0, 2], [2, 4]\}$.
- $\lambda^1_{n,j} = 1$ if $x_{n,j} \in [0, 2]$;
- $\lambda^2_{n,j} = 1$ if $x_{n,j} \in [2, 4]$.

Logical Constraints

Step 3: Logical Constraints

- Ensure each $x_{n,j}$ is assigned to exactly one interval:

$$\sum_{\sigma \in \mathcal{S}_{n,j}} \lambda_{n,j}^{\sigma} = 1.$$

- Enforce consistency between $x_{n,j}$ and $\lambda_{n,j}^{\sigma}$:

$$\sum_{\sigma \in \mathcal{S}_{n,j}} \underline{x}_{n,j}^{\sigma} \lambda_{n,j}^{\sigma} \leq x_{n,j} \leq \sum_{\sigma \in \mathcal{S}_{n,j}} \overline{x}_{n,j}^{\sigma} \lambda_{n,j}^{\sigma}.$$

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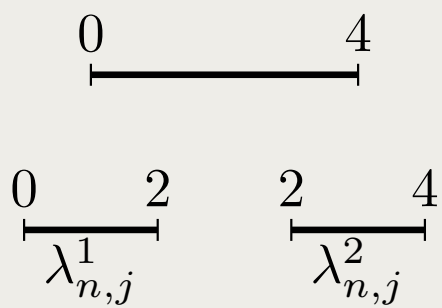
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$$\lambda_{n,j} \in \{0,1\}^2$$

$$\lambda_{n,j}^1 + \lambda_{n,j}^2 = 1$$

$$0 \cdot \lambda_{n,j}^1 + 2 \cdot \lambda_{n,j}^2 \leq x_{n,j} \leq 2 \cdot \lambda_{n,j}^1 + 4 \cdot \lambda_{n,j}^2$$

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Lift the x_n space to x_n-λ_n space:

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In the next few slides, I will explain:

- How to approximate Q_m in the lifted space;
- Why lifting enables tighter, piecewise convex-envelope approximations.

Cutting-plane Approximation

In x_n-λ_n space, the cut approximator is defined as:

$$\phi_m^L(x_n, \lambda_n) := \min \left\{ \theta_m \;\middle|\; \theta_m \geq v_m^\ell + {\pi_m^\ell}^\top x_n + {\rho_m^\ell}^\top \lambda_n, \quad \forall \ell = 0, \ldots, L_m \right\}.$$

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Define $\underline{Q}_n^L(x_{a(n)}, \lambda_{a(n)}; \Phi_n^L, \mathbb{P}_n, \mathbb{P}_{a(n)}) :=$

$$\begin{aligned} \min \quad & c_n^\top x_n + g_n^\top y_n + \sum_{m \in \mathcal{C}(n)} q_{nm} \cancel{Q_m(x_n)} \phi_m^L(x_n, \lambda_n) && (P_n^L) \\ \text{s.t.} \quad & W_n x_n + B_n y_n = T_n z_n + d_n \\ & x_n \in X_n, y_n \in Y_n, \lambda_n \in \Lambda_n(x_n, \mathbb{P}_n) \\ & z_n = x_{a(n)}, \mu_n = \lambda_{a(n)}. && \text{(Non-Anticipativity)} \end{aligned}$$

- $\lambda_n \in \Lambda_n(x_n, \mathbb{P}_n)$ ensures that λ_n are consistent with the partitions $\mathbb{P}_n$.

Lifting the Lagrangian Dual Problem

Given: A lifted state $(\hat{x}_n, \hat{\lambda}_n)$ at node n.

Define the lifted Lagrangian dual problem at node $m \in \mathcal{C}(n)$ as:

$$\max_{\pi_m, \rho_m} \quad \mathcal{L}_m^L(\pi_m, \rho_m; \hat{x}_n, \hat{\lambda}_n, \Phi_m^L, \mathcal{Z}_n, \mathbb{P}_m, \mathbb{P}_n), \tag{D_m^L}$$

where $\mathcal{L}_m^L(\pi_m, \rho_m; \hat{x}_n, \hat{\lambda}_n, \Phi_m^L, \mathcal{Z}_n, \mathbb{P}_m, \mathbb{P}_n) :=$

$$\begin{aligned} \min \quad & c_m^\top x_m + g_m^\top y_m + \sum_{\tau \in \mathcal{C}(m)} q_{m\tau} \phi_\tau^L(x_m, \lambda_m) + \\ & \qquad \pi_m^\top (\hat{x}_n - z_m) + \rho_m^\top (\hat{\lambda}_n - \mu_m) \\ \text{s.t.} \quad & W_m x_m + B_m y_m = T_m z_m + d_m \\ & x_m \in X_m, y_m \in Y_m, \lambda_m \in \Lambda_m(x_m, \mathbb{P}_m) \\ & z_m \in \mathcal{Z}_n, \mu_m \in \Lambda_n(z_m, \mathbb{P}_n) \end{aligned}$$

Lifted Lagrangian Cut

Given: An optimal dual solution (π_m^*, ρ_m^*) to (D_m^L).

Generate a lifted Lagrangian cut:

$$\theta_m \geq v_m^* + {\pi_m^*}^\top x_n + {\rho_m^*}^\top \lambda_n.$$

where $v_m^* = \mathcal{L}_m^L(\pi_m^*, \rho_m^*) - {\pi_m^*}^\top \hat{x}_n - {\rho_m^*}^\top \hat{\lambda}_n$.

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What are the properties of the lifted Lagrangian cuts?

Properties of Lifted Lagrangian Cuts

Define the restricted region in the lifted space (counterpart of $\mathcal{Z}_n$):

$$\mathfrak{Z}_n(\mathcal{Z}_n, \mathbb{P}_n) := \{(x_n, \lambda_n) \mid x_n \in \mathcal{Z}_n, \lambda_n \in \Lambda_n(x_n, \mathbb{P}_n)\}.$$

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Then the lifted Lagrangian cut

$$\theta_m \geq v_m^* + (\pi_m^*)^\top x_n + (\rho_m^*)^\top \lambda_n$$

is:

1. **valid** over $\mathfrak{Z}_n(\mathcal{Z}_n, \mathbb{P}_n)$, i.e., $\forall (x_n, \lambda_n) \in \mathfrak{Z}_n(\mathcal{Z}_n, \mathbb{P}_n)$

$$Q_m(x_n) \geq \underline{Q}_m^L(x_n, \lambda_n; \Phi_m^L, \mathbb{P}_m, \mathbb{P}_n) \geq v_m^* + (\pi_m^*)^\top x_n + (\rho_m^*)^\top \lambda_n;$$

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2. **tight** for the convex envelope over $\mathfrak{Z}_n(\mathcal{Z}_n, \mathbb{P}_n)$ at incumbent $(\hat{x}_n, \hat{\lambda}_n)$:

$$\mathcal{L}_m^L(\pi_m^*, \rho_m^*) = \mathrm{co}_{\mathfrak{Z}_n(\mathcal{Z}_n, \mathbb{P}_n)} \left(\underline{Q}_m^L(\cdot, \cdot; \Phi_m^L, \mathbb{P}_m, \mathbb{P}_n) \right) (\hat{x}_n, \hat{\lambda}_n);$$

3. **tight** in original space:

   The active region for the lifted state $(\hat{x}_n, \hat{\lambda}_n)$ is:

$$\hat{\mathcal{Z}}_n := \mathcal{Z}_n \cap \prod_j \left[\underline{x}_{n,j}^{\hat{\sigma}_j},\ \overline{x}_{n,j}^{\hat{\sigma}_j}\right] \text{ where } \hat{\lambda}_{n,j}^{\hat{\sigma}_j} = 1,$$

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   We recover:

$$\mathcal{L}_m^L(\pi_m^*, \rho_m^*) = \mathrm{co}_{\hat{\mathcal{Z}}_n}\left(\underline{Q}_m(\cdot; \Phi_m^L)\right)(\hat{x}_n),$$

   i.e., cut value equals the convex envelope of $\underline{Q}_m(\cdot; \Phi_m^L)$ over $\hat{\mathcal{Z}}_n$.

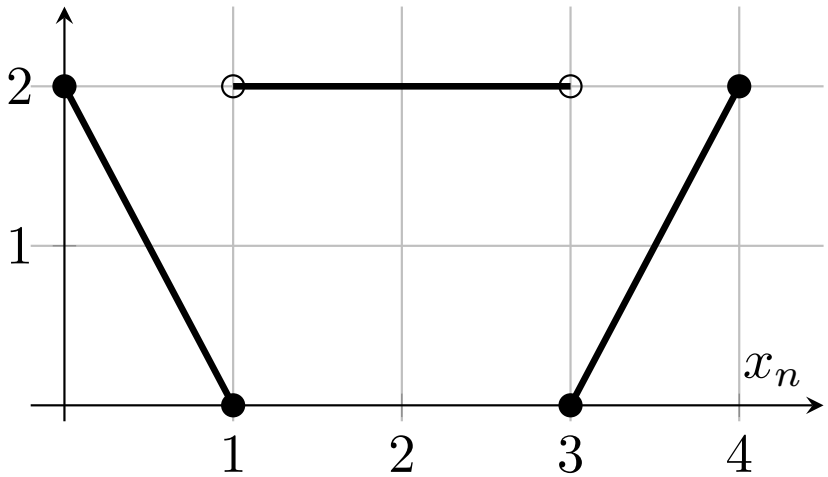
This explains why lifting can achieve recovering the piecewise convex envelope.

Example – Convex Envelope

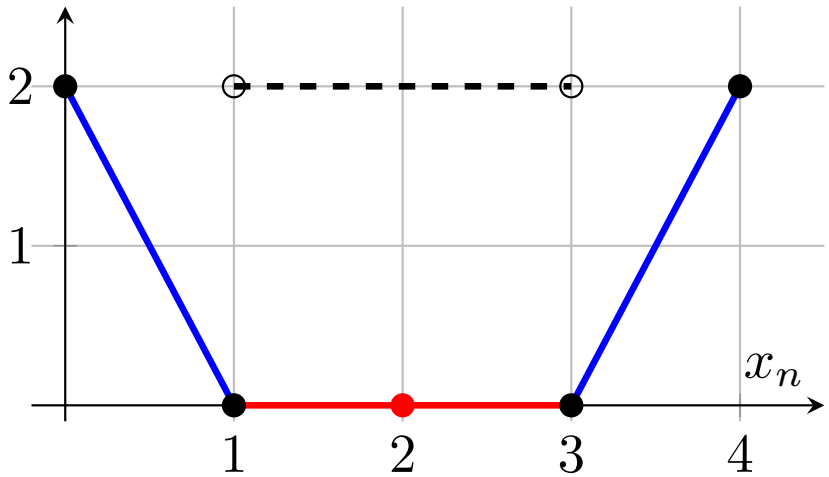
Let us consider the value function

$$Q_m(x_n) = \begin{cases} -2x_n + 2 & \text{if } x_n \in [0,1] \\ 2 & \text{if } x_n \in (1,3) \\ 2x_n - 6 & \text{if } x_n \in [3,4]. \end{cases}$$

Approximation using Lagrangian cuts with $\mathcal{Z}_n = [0,4]$:
(i) tight in the blue regions, (ii) with a gap in the red region.



(a) Value function Q_m



(b) Convex envelope of Q_m over $\mathcal{Z}_n$

Example – Lifting

Since $x_n \in \mathbb{R}$ is continuous over $[0, 4]$, we split $[0, 4]$ at $\hat{x}_n = 2$:



$\mathcal{P}_n = \{[0, 2], [2, 4]\}$.

The lifted state variable is (x_n, λ_n).



$\mathbb{P}_n$: $\lambda_n = 0$ if $x_n \in [0, 2]$;
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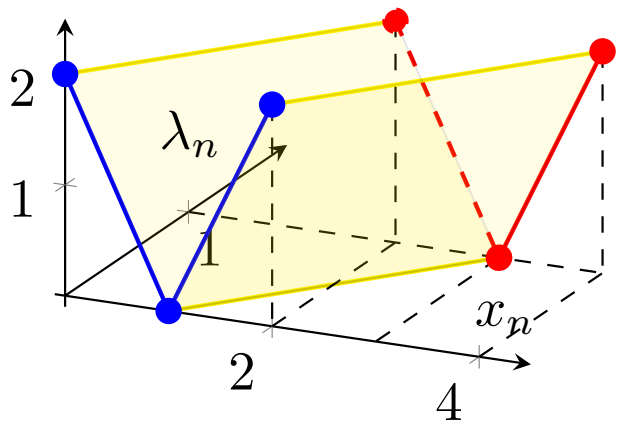
Lifted states $(x_n, \lambda_n) = (2, 1)$ & $(2, 0)$ are extreme points of the set

$$\mathfrak{Z}_n(\mathcal{Z}_n, \mathbb{P}_n) = \left\{ (x_n, \lambda_n) \;\middle|\; \begin{array}{l} \lambda_n \in \{0, 1\} \\ 0 \cdot (1 - \lambda_n) + 2 \cdot \lambda_n \leq x_n \leq 2 \cdot (1 - \lambda_n) + 4 \cdot \lambda_n \end{array} \right\}.$$

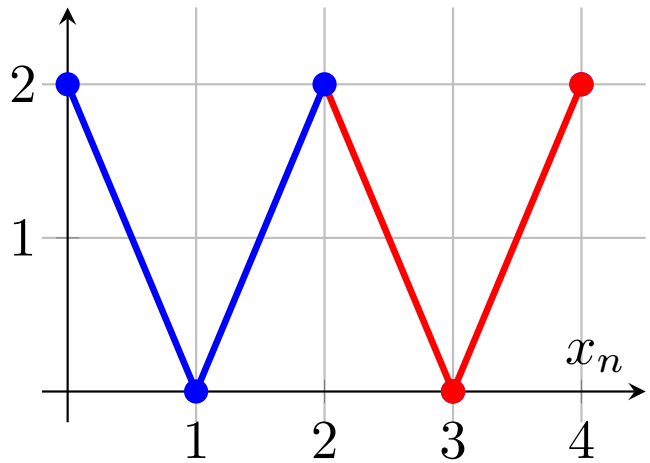
Example – Lifted Convex Envelope

Approximate the value function Q_m via lifted Lagrangian cuts:

$$\theta_m \geq -2x_n + 4\lambda_n + 2,$$
$$\theta_m \geq 2x_n - 4\lambda_n - 2.$$



(c) Convex envelope of in x_n-λ_n space.



(d) Piecewise convex envelope in x_n space.

Figure: Geometric representation of cut approximators.

Stochastic Dual Dynamic Programming with Lifting (SDDP-L) Algorithm

Partition Refinement

Objective: In each iteration, dynamically

- Lift state-variable space;
- Tighten piecewise convex envelope.

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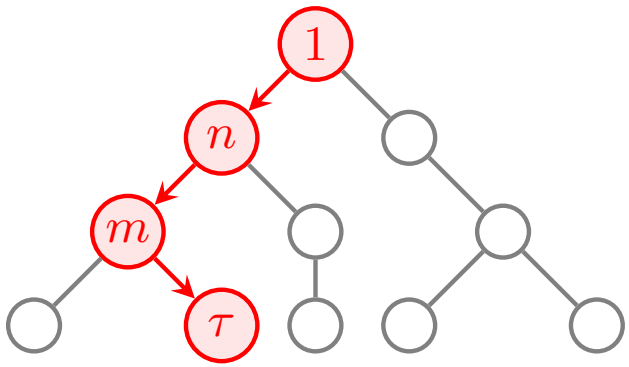
Refinement Ideas: For active interval $[\underline{x}^{\hat{\sigma}}_{n,j}, \overline{x}^{\hat{\sigma}}_{n,j}]$ of component $x_{n,j}$:

- Split at incumbent solution: $[\underline{x}^{\hat{\sigma}}_{n,j}, \hat{x}_{n,j}]$ and $[\hat{x}_{n,j}, \overline{x}^{\hat{\sigma}}_{n,j}]$;
- Split at mid-point: $[\underline{x}^{\hat{\sigma}}_{n,j}, \frac{\underline{x}^{\hat{\sigma}}_{n,j}+\overline{x}^{\hat{\sigma}}_{n,j}}{2}]$ and $[\frac{\underline{x}^{\hat{\sigma}}_{n,j}+\overline{x}^{\hat{\sigma}}_{n,j}}{2}, \overline{x}^{\hat{\sigma}}_{n,j}]$.

SDDP-L – Forward Pass

Forward Pass of Iteration i:

- Sample a scenario path $\mathcal{P}^{i,\omega} = \{1, n, m, \tau\}$;



- **For** each node $n \in \mathcal{P}^{i,\omega}$:

  Solve problem $(P_n^{L,i})$:

$$\min \quad c_n^\top x_n + g_n^\top y_n + \sum_{m \in \mathcal{C}(n)} q_{nm} \phi_m^{L,i}(x_n, \lambda_n) \qquad (P_n^{L,i})$$

$$\text{s.t.} \quad W_n x_n + B_n y_n = T_n z_n + d_n$$

$$x_n \in X_n, y_n \in Y_n, \lambda_n \in \Lambda_n(x_n, \mathbb{P}_n^i)$$

$$z_n = x_{a(n)}^i, \mu_n = \lambda_{a(n)}^i.$$

SDDP-L – Backward Pass

Backward Pass of Iteration i:

- **For** each node $n \in \mathcal{P}^{i,\omega}$:

  Refine partition $\mathbb{P}_n^i$ to obtain $\mathbb{P}_n^{i+1}$;

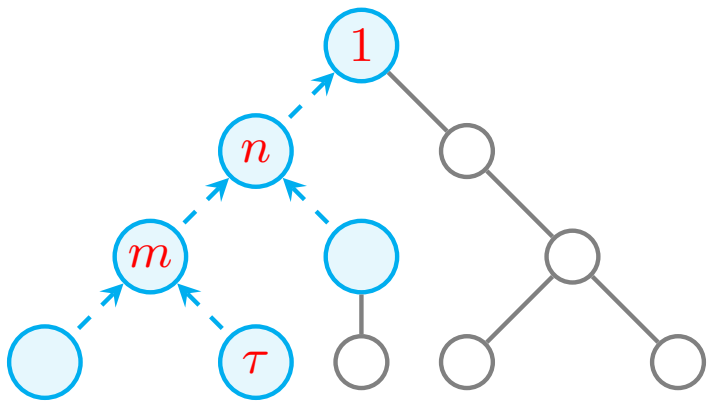
  - **For** each child node $m \in \mathcal{C}(n)$:

    Solve the Lagrangian dual problem $(D_m^{L,i})$

$$\max_{\pi_m, \rho_m} \quad \mathcal{L}_m^L(\pi_m, \rho_m; x_n^i, \lambda_n^i, \Phi_m^{L,i+1}, \mathcal{Z}_n, \mathbb{P}_m^{i+1}, \mathbb{P}_n^{i+1}), \qquad (D_m^{L,i})$$

    to generate a cut of the form

$$\theta_m \geq v_m + \pi_m^\top x_n + \rho_m^\top \lambda_n.$$



Global Convergence

Assumptions: For every node $n \in \mathcal{N}$:

- The size of every element in partition $\mathbb{P}_n^i$ diminishes to 0, i.e.,

$$(\overline{x}_{n,j}^{\sigma,i} - \underline{x}_{n,j}^{\sigma,i}) \overset{i\to\infty}{\longrightarrow} 0, \ \forall j, \sigma.$$

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Theorem

The SDDP-L algorithm generates a sequence of optimal solutions $\left\{\left(x_1^i, y_1^i\right)\right\}_{i\in\mathbb{N}}$ *to the root node problem* (P_n^L). *With probability one, every accumulation point* (x_1^*, y_1^*) *of* $\left\{\left(x_1^i, y_1^i\right)\right\}_{i\in\mathbb{N}}$ *is an optimal solution to model* (P), *and the lower bounds converge to the optimal value* V^*.

Enhanced Lagrangian Cuts

"Curse" of Extreme Points

Let $\Pi_m(\hat{x}_n)$ be the set of optimal solutions to (D_m).

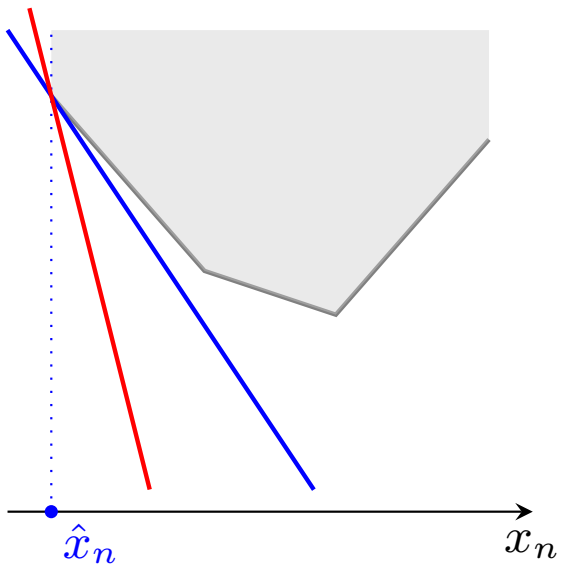
$$\max_{\pi_m} \quad \mathcal{L}_m\left(\pi_m; \hat{x}_n, \Phi_m, \mathcal{Z}_n\right) \qquad (D_m)$$

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Any $\pi_m \in \Pi_m(\hat{x}_n)$ gives a tight cut:

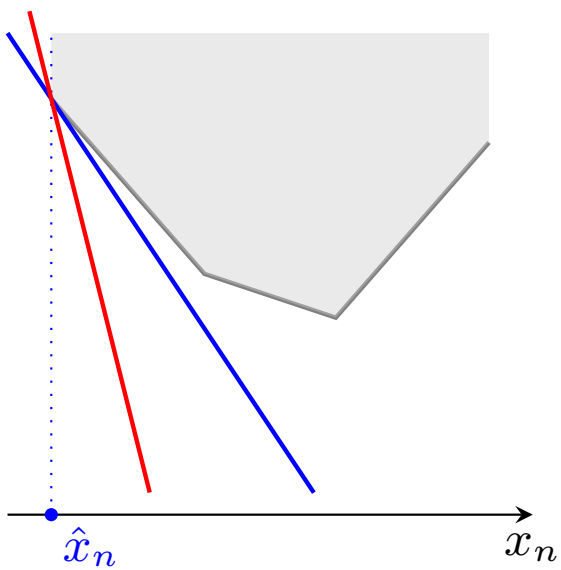


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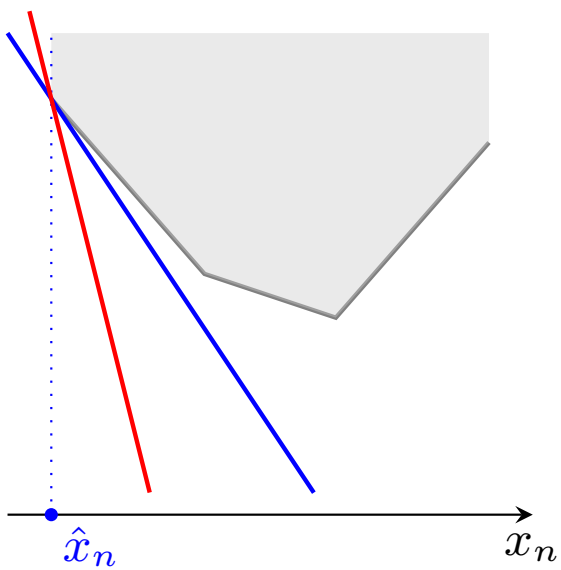
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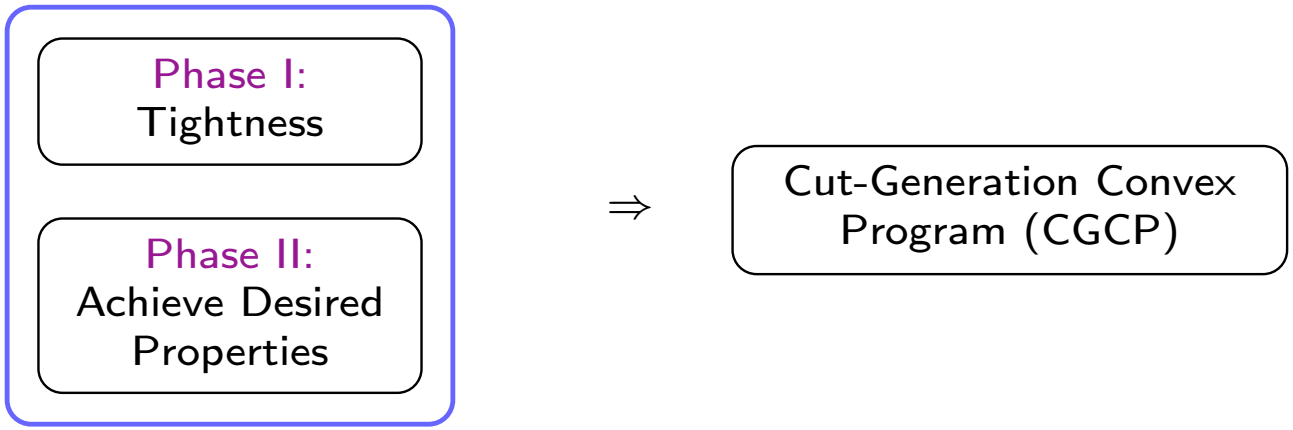


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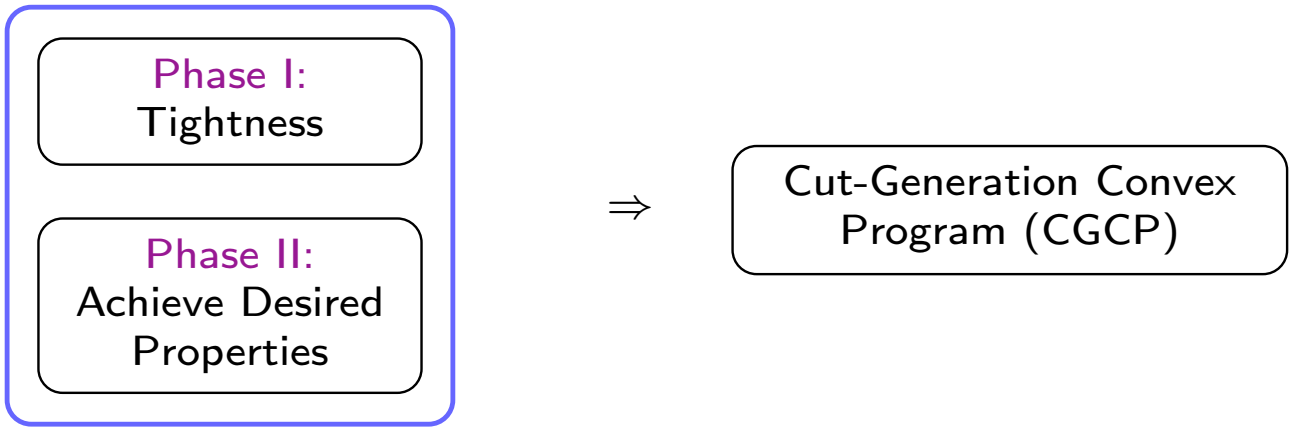
Can we make sure that:

- at $\hat{x}_n$, the cut recovers the convex envelope's value;
- at somewhere else, the cut will "push" the value up?

Two-Phase Ehancement Method



Two-Phase Ehancement Method

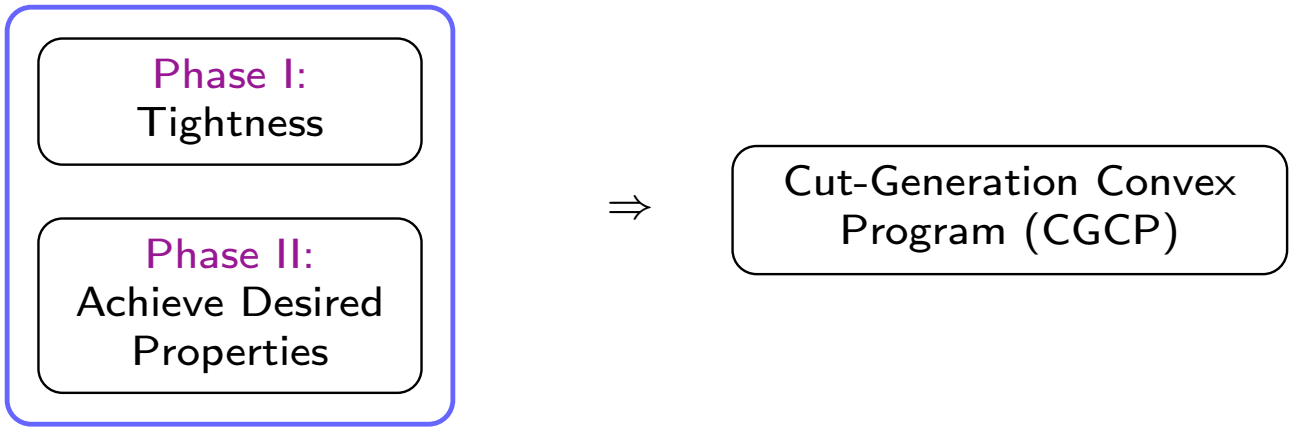


Phase I: Tightness at the Incumbent Point

The **tightness** at $\hat{x}_n$ can be achieve by enforcing:

$$\pi_m \in \Pi_m(\hat{x}_n). \tag{T}$$

Two-Phase Ehancement Method



Phase I: Tightness at the Incumbent Point

The **tightness** at $\hat{x}_n$ can be achieve by enforcing:

$$\pi_m \in \Pi_m(\hat{x}_n). \tag{T}$$

Convex relaxation: *Given* $\hat{\pi}_m \in \Pi_m(\hat{x}_n)$, and a small tolerance $\epsilon > 0$.

To achieve convexity, replace (T) above with

$$\mathcal{L}_m\left(\pi_m; \hat{x}_n, \Phi_m, \mathcal{Z}_n\right) \geq \mathcal{L}_m\left(\hat{\pi}_m; \hat{x}_n, \Phi_m, \mathcal{Z}_n\right) - \epsilon.$$

Pareto-optimal Lagrangian Cut (PLC)

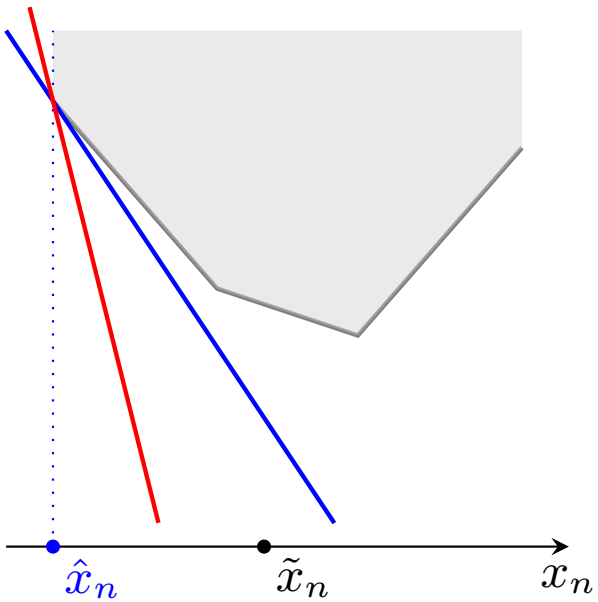
Phase II: Pareto-optimality

Phase II: Improve Cut Values at Core Point

Let $\tilde{x}_n \in \text{relint}(\text{conv}(\mathcal{Z}_n))$ be a core point.

- **Maximize** its cut value at $\tilde{x}_n$:

$$\max \quad \mathcal{L}_m\left(\pi_m; \tilde{x}_n, \Phi_m, \mathcal{Z}_n\right)$$



Pareto-optimal Lagrangian Cut (PLC)

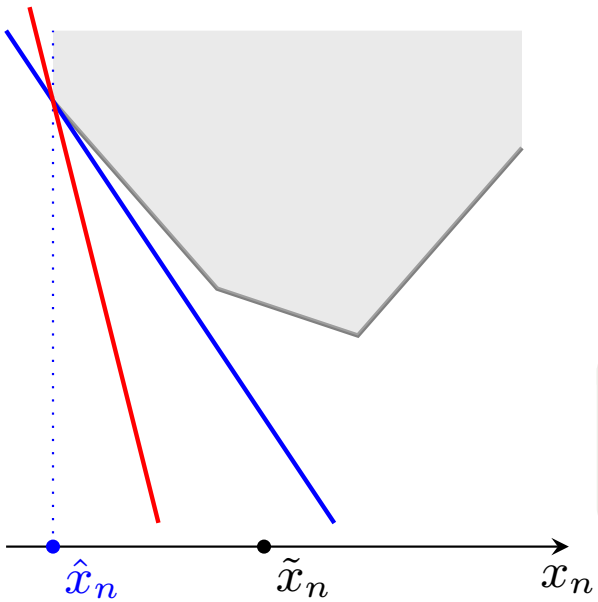
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- **Maximize** its cut value at $\tilde{x}_n$:

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$$\begin{aligned} \max_{\pi_m} \quad & \mathcal{L}_m\left(\pi_m; \hat{x}_n, \Phi_m, \mathcal{Z}_n\right) + \pi_m^\top(\tilde{x}_n - \hat{x}_n) \\ \text{s.t.} \quad & \mathcal{L}_m\left(\pi_m; \hat{x}_n, \Phi_m, \mathcal{Z}_n\right) \geq \\ & \quad \mathcal{L}_m\left(\hat{\pi}_m; \hat{x}_n, \Phi_m, \mathcal{Z}_n\right) - \epsilon. \end{aligned}$$

PLC is Pareto-optimal.

Square-Minimization Cut (SMC)

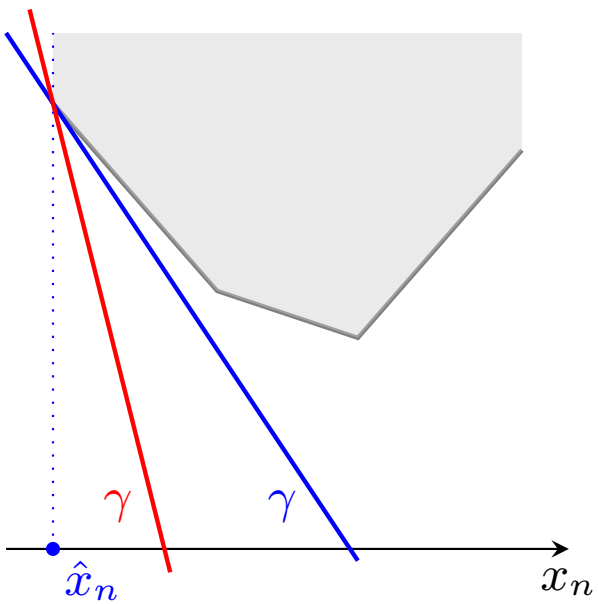
Phase II: Flatness

Phase II: Generate "Flat" Cuts

Let γ be the angle formed by the cut and the horizontal axis.

- **Maximize** cosine of the angle γ:

$$\max \quad \cos(\gamma) := \frac{1}{\sqrt{{\pi_m}^\top \pi_m + 1}}$$



Square-Minimization Cut (SMC)

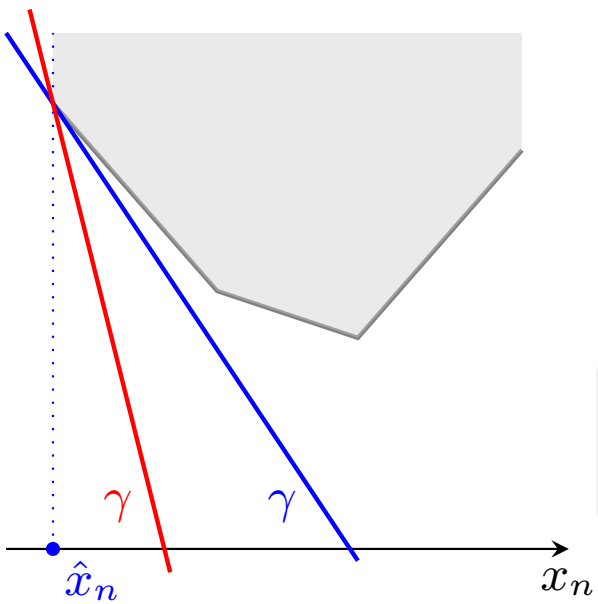
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$$\begin{aligned} \min_{\pi_m} \quad & \pi_m^\top \pi_m \\ \text{s.t.} \quad & \mathcal{L}_m\left(\pi_m; \hat{x}_n, \Phi_m, \mathcal{Z}_n\right) \geq \\ & \qquad \mathcal{L}_m\left(\hat{\pi}_m; \hat{x}_n, \Phi_m, \mathcal{Z}_n\right) - \epsilon. \end{aligned}$$

SMC avoids selecting a core point.

Numerical Experiments

Multistage Stochastic Unit Commitment (MSUC)

- MSUC with power flow, power balance, ramping, and minimum-up/-down constraints to minimize production and load-shedding costs.

$$\min \quad \sum_{t\in\mathcal{T}} \left[\sum_{g\in\mathcal{G}} \left(h_{gt} + c_g^{Start} v_{gt} + c_g^{Down} w_{gt} \right) + \sum_{d\in\mathcal{D}} w_d (1 - x_{dt}) \right]$$

$$\text{s.t.} \quad h_{gt} \geq c_{og}^1 P_{gt}^G + c_{og}^0 y_{gt} \qquad \forall o \in \mathcal{O}, g \in \mathcal{G}, t \in \mathcal{T}$$

$$P_{ijt}^L = B_{ij} \left(\theta_{it} - \theta_{jt} \right) \qquad \forall (i,j) \in \mathcal{L}, t \in \mathcal{T}$$

$$-W_{ij} \leq P_{ijt}^L \leq W_{ij} \qquad \forall (i,j) \in \mathcal{L}$$

$$\underline{P}_g^G y_{gt} \leq P_{gt}^G \leq \overline{P}_g^G y_{gt} \qquad \forall g \in \mathcal{G}, t \in \mathcal{T}$$

$$\sum_{g\in\mathcal{G}_i} P_{gt}^G + \sum_{(i,j)\in\mathcal{L}} P_{ijt}^L = \sum_{d\in\mathcal{D}_i} D_{dt} x_{dt} \qquad \forall i \in \mathcal{B}, t \in \mathcal{T}$$

$$v_{gt} - w_{gt} = y_{gt} - y_{g,t-1} \qquad \forall g \in \mathcal{G}, t \in \mathcal{T}$$

$$P_{gt}^G - P_{g,t-1}^G \leq M_g y_{g,t-1} + \underline{P}_g^G v_{gt} \qquad \forall g \in \mathcal{G}, t \in \mathcal{T}$$

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$$v_{gt}, y_{gt}, w_{gt} \in \{0,1\} \qquad \forall g \in \mathcal{G}, t \in \mathcal{T}$$

$$x_{dt} \in [0,1] \qquad \forall d \in \mathcal{D}, t \in \mathcal{T}$$

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$$x_{dt} \in [0,1] \qquad \forall d \in \mathcal{D}, t \in \mathcal{T}$$

- **State Variables**: Binary v, y, w & Continuous P^G.

Experiment – Setup

We consider three algorithms

- SDDP-L: SDDP + Lifted Lagrangian cuts;
- SDDiP: binarize all non-binary variables $\rightarrow$ Approximation;
- SDDP: Lagrangian cuts.

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- SDDP-L: SDDP + Lifted Lagrangian cuts;
- SDDiP: binarize all non-binary variables $\rightarrow$ Approximation;
- SDDP: Lagrangian cuts.

With the following setups:

- Use IEEE case30 [Shahidehpour and Wang, 2003];
- Random variables: the demand D_{dt};
- CGCPs are solved by using the level method [Lemaréchal et al., 1995];

$\rightsquigarrow$ Time Limit $= 3,600$s.

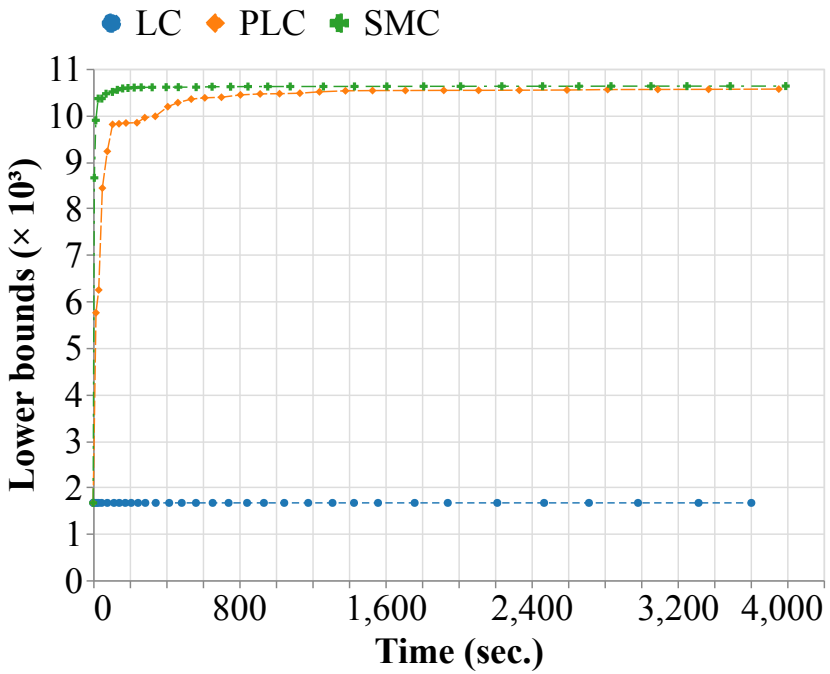
SDDP-L – Incumbent Solutions

SDDP-L Algorithm with Partitions Refined according to Incumbent Solutions								
Cut	T	R	LB	Gap(%)	Iter.	Time/Iter.	$\text{Time}_{1\%}$	$\text{Iter}_{1\%}$
LC	6	5	1629.5	61.3	52	73.4 (54.1)	-	-
		10	1296.2	66.4	51	74.5 (55.9)	-	-
	8	5	1662.8	74.1	44	88.3 (68.4)	-	-
		10	1662.8	74.2	43	88.8 (70.1)	-	-
	12	5	1662.8	84.5	33	112.6 (94.0)	-	-
		10	1999.7	83.2	45	85.4 (44.9)	-	-
PLC	6	5	4203.6	0.1	56	66.3 (55.0)	24.3	4
		10	3849.0	0.2	48	79.2 (57.0)	33.1	4
	8	5	6405.7	0.3	42	88.0 (70.9)	85.7	8
		10	6386.8	0.6	39	96.1 (62.5)	290.1	10
	12	5	10563.1	1.0	32	121.1 (94.3)	1709.9	24
		10	11574.1	2.5	30	133.4 (89.1)	-	-
SMC	6	5	4206.3	0.1	57	66.0 (41.4)	16.6	4
		10	3853.4	0.0	53	71.5 (49.4)	17.3	4
	8	5	6413.1	0.1	44	87.9 (60.8)	46.4	6
		10	6399.3	0.0	43	88.8 (63.6)	71.5	7
	12	5	10627.8	0.1	34	114.9 (84.8)	161.7	9
		10	11749.2	0.5	36	104.8 (80.4)	308.3	12

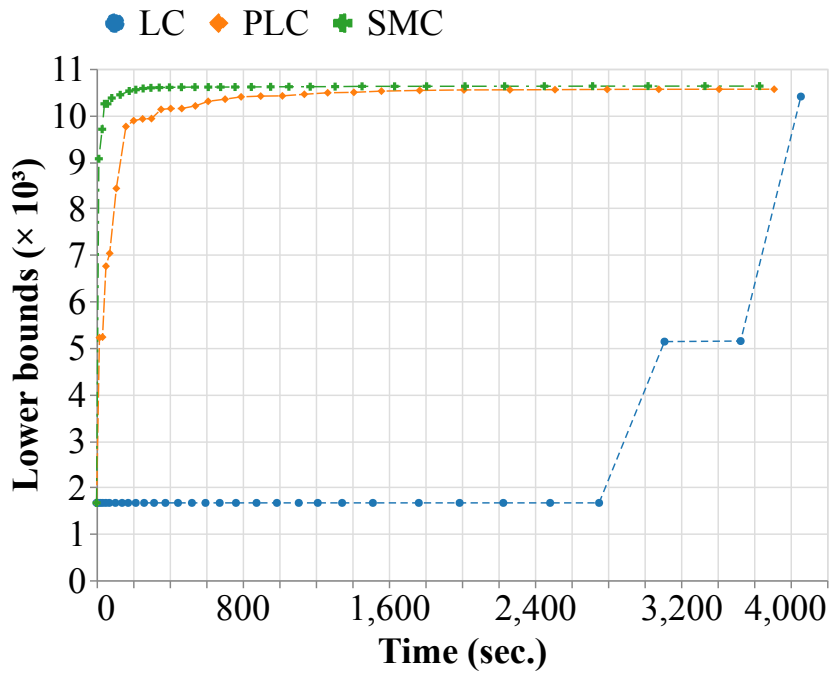
SDDP-L – Bisection

SDDP-L Algorithm with Partitions Refined according to Bisection

Cut	T	R	LB	Gap(%)	Iter.	Time/Iter.	$\text{Time}_{1\%}$	$\text{Iter}_{1\%}$
LC	6	5	4137.5	1.7	50	77.2 (60.6)	-	-
		10	3788.2	1.7	48	77.5 (60.5)	-	-
	8	5	6342.2	0.9	42	91.8 (80.9)	2082.2	35
		10	4004.9	37.4	41	94.5 (81.5)	-	-
	12	5	10402.3	10.8	32	124.4 (112.7)	-	-
		10	1999.7	83.0	32	121.0 (94.0)	-	-
PLC	6	5	4203.4	0.1	49	76.2 (55.4)	30.7	4
		10	3847.6	0.2	46	83.2 (55.8)	30.2	4
	8	5	6408.8	0.4	41	92.1 (76.3)	190.6	10
		10	6389.5	0.2	39	101.5 (78.0)	146.0	8
	12	5	10560.3	1.0	31	123.7 (97.8)	2796.1	28
		10	11568.1	1.9	28	133.6 (99.6)	-	-
SMC	6	5	4206.4	0.1	49	75.0 (64.8)	18.0	4
		10	3853.1	0.1	50	74.3 (59.2)	18.5	4
	8	5	6413.7	0.3	42	91.6 (75.1)	73.9	7
		10	6399.2	0.5	41	91.2 (78.7)	129.1	9
	12	5	10627.1	0.0	33	113.4 (92.1)	254.9	10
		10	11746.0	0.8	32	117.7 (96.2)	256.7	10



(a) $R = 5$, SDDP-L, refining partition according to incumbent solutions.



(b) $R = 5$, SDDP-L, refining partition according to bisection.

Figure: Progression of lower bounds for the MSUC instance with T = 12.

Benchmark vs. Binarization

SDDiP Algorithm with State Variables Binary Representation of Precision 2^{-8}								
Cut	T	R	LB	Gap(%)	Iter.	Time/Iter.	$\text{Time}_{1\%}$	$\text{Iter}_{1\%}$
LC	6	5	2037.9	54.6	30	129.5 (67.3)	-	-
		10	1554.4	61.7	32	123.3 (77.0)	-	-
	8	5	2003.7	69.5	29	138.0 (72.2)	-	-
		10	2003.4	69.8	25	159.0 (90.1)	-	-
	12	5	1952.8	82.5	24	160.6 (63.2)	-	-
		10	2395.9	80.4	22	180.4 (72.9)	-	-
PLC	6	5	2832.3	35.0	7	704.5 (114.2)	-	-
		10	2391.1	41.5	5	983.4 (125.7)	-	-
	8	5	2483.1	64.4	6	866.2 (127.0)	-	-
		10	2275.5	65.6	4	1569.0 (322.4)	-	-
	12	5	2271.1	79.4	4	1217.3 (35.9)	-	-
		10	2705.5	78.3	3	1900.9 (497.8)	-	-
SMC	6	5	4266.1	0.1	60	62.5 (29.0)	120.4	5
		10	3887.9	0.2	30	127.2 (48.9)	349.6	5
	8	5	5354.9	18.2	23	171.2 (55.7)	-	-
		10	5437.8	16.6	23	173.4 (76.3)	-	-
	12	5	10041.1	8.4	21	193.0 (68.7)	-	-
		10	10985.2	8.5	27	149.0 (82.6)	-	-

Duality Gap Closed via Lifting

SDDP with Lagrangian Cuts (Algorithm 1)						
Cut	T	R	LB	Gap(%)	Iter.	Time/Iter.
LC	6	5	1629.5	61.3	116	31.4 (24.3)
		10	1296.2	66.4	108	34.3 (21.5)
	8	5	1662.8	74.1	94	39.9 (30.0)
		10	1662.8	74.1	97	37.5 (26.0)
	12	5	1662.8	84.4	72	51.9 (44.6)
		10	1999.7	83.1	76	49.2 (32.1)
PLC	6	5	4169.3	0.9	108	34.4 (25.2)
		10	3819.0	0.9	89	41.1 (22.7)
	8	5	6327.1	1.7	89	41.6 (28.9)
		10	6277.8	2.1	82	45.6 (28.2)
	12	5	10476.0	2.0	65	56.5 (44.0)
		10	11574.4	1.5	62	59.2 (34.6)
SMC	6	5	4172.7	0.9	131	27.8 (21.2)
		10	3822.6	0.8	104	35.8 (29.8)
	8	5	6330.3	1.6	104	35.0 (25.3)
		10	6265.6	2.0	94	39.2 (27.9)
	12	5	10490.3	1.7	80	46.8 (39.0)
		10	11571.9	1.9	70	52.9 (45.3)

Outline

1 Introduction

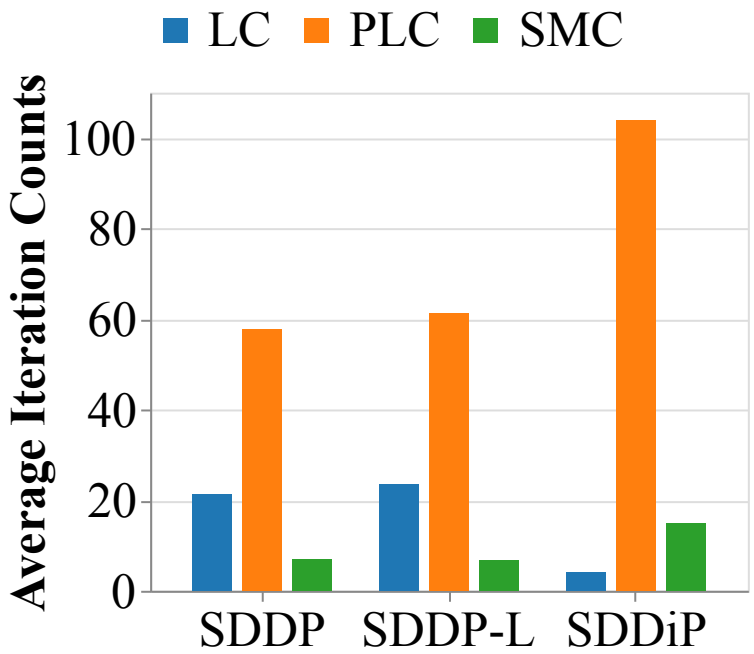
2 Lift-and-Cut Algorithm

3 Improving Benders Cuts via Disjunctive Programming

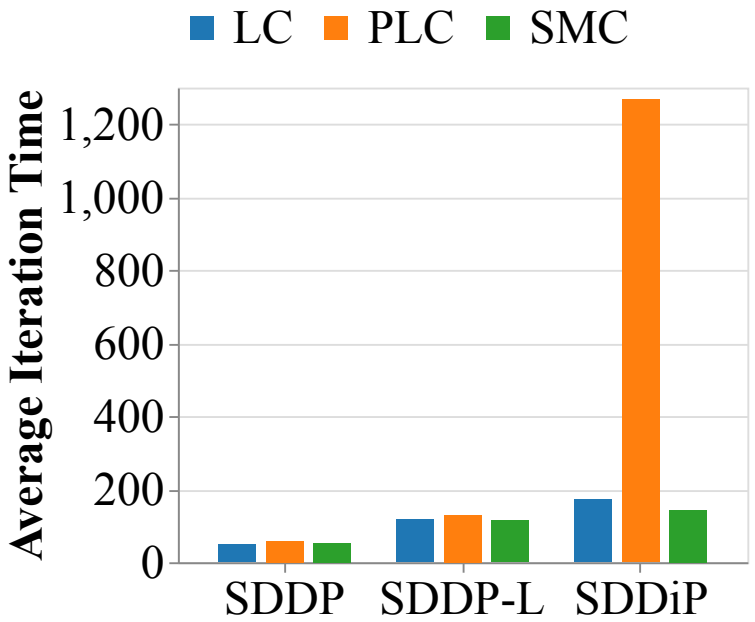
4 Two-stage Model for Mitigating Wildfire Risk

5 Summary

Limitation of Solving CGCPs



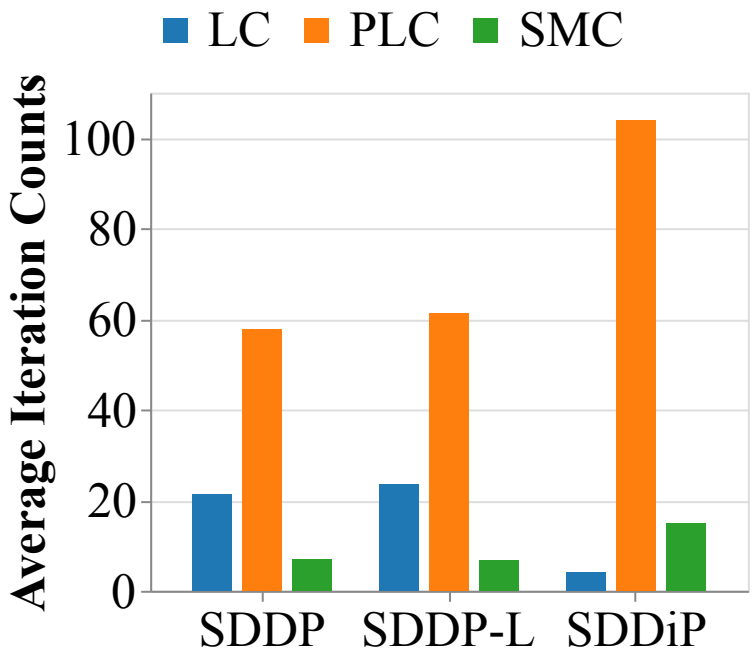
(a) Average number of iterations in the level method required to generate a Lagrangian cut.



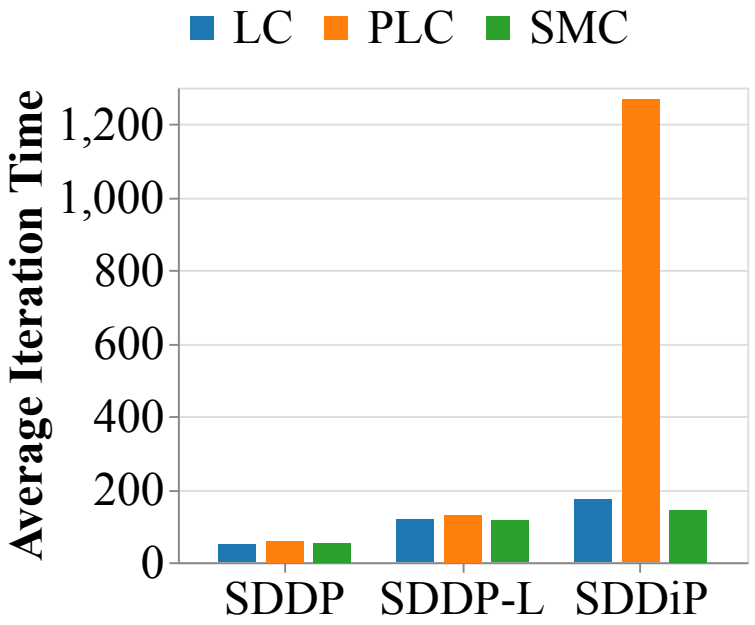
(b) Average time per iteration of the level method to generate a Lagrangian cut.

Figure: Statistics on CGCPs for the MSUC instance with T = 12 and R = 10.

Limitation of Solving CGCPs



(a) Average number of iterations in the level method required to generate a Lagrangian cut.



(b) Average time per iteration of the level method to generate a Lagrangian cut.

Figure: Statistics on CGCPs for the MSUC instance with T = 12 and R = 10.

Question: Can we exploit special structure to replace CGCPs with CGLPs?

Nodal subproblem at node $m \in \mathcal{N}$:

$$Q_m(x_n) :=$$

$$\min_{x_m, y_m} \quad c_m^\top x_m + g_m^\top y_m + \overbrace{\sum_{\tau \in \mathcal{C}(m)} q_{m\tau} Q_\tau(x_m)}^{\mathcal{Q}_m(x_m) :=}$$

$$\text{s.t.} \quad W_m x_m + B_m y_m = T_m x_n + d_m$$

$$x_m \in X_m = \{0,1\}^{\dim(x_m)}, \; y_m \in Y_m$$

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- Applications with binary states:
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- Approximate $\mathcal{Q}_m(x_m)$ using an aggregated cut approximator:

$$\mathfrak{Q}_m^i(x_m) := \min \left\{ \theta_m \mid \theta_m \geq \overline{v}_m^\ell + (\overline{\lambda}_m^\ell)^\top x_m, \; \forall \ell = 0, 1, \ldots, i \right\}.$$

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- Define $\mathfrak{X}_m^i$ as the constraint system:

$$\mathfrak{X}_m^i := \left\{ (z_m, x_m, y_m, \theta_n) \;\middle|\; \begin{array}{l} W_m x_m + B_m y_m = T_m z_m + d_m \\ x_m \in X_m, \quad y_m \in Y_m \\ \theta_m \geq \overline{v}_m^\ell + (\overline{\lambda}_m^\ell)^\top x_m, \quad \forall \ell = 0, 1, \ldots, i \end{array} \right\}.$$

Convex Hull Reformulation

- An under-approximation of Q_m is:

$$\begin{aligned} Q_m(x_n^i; \mathfrak{Q}_m^i) := \min \quad & c_m^\top x_m + g_m^\top y_m + \theta_m \\ \text{s.t.} \quad & (z_m, x_m, y_m, \theta_m) \in \mathfrak{X}_m^i \qquad (P_m(x_n^i; \mathfrak{Q}_m^i)) \\ & z_m = x_n^i. \end{aligned}$$

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Limitation

The convex region depends on x_n:

$\operatorname{conv}\left(\mathfrak{X}_m^i \wedge \{z_m = x_n\}\right)$ is **parameterized** by x_n

Key Idea: Introducing $\mathcal{Z}_n$

Introduce a set $\mathcal{Z}_n$ to capture all possible values of $x_n \in \{0,1\}^d$:

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Theorem

Let $x_n^i \in \{0,1\}^d$, and let $\mathcal{Z}_n = [0,1]^d$ or $\{0,1\}^d$. Then:

$$\text{conv}\left(\mathfrak{X}_m^i \wedge \{z_m \in \mathcal{Z}_n\}\right) \wedge \{z_m = x_n^i\} = \text{conv}\left(\mathfrak{X}_m^i \wedge \{z_m \in \mathcal{Z}_n\} \wedge \{z_m = x_n^i\}\right)$$

Convex-Hull Relaxation ⇒ Tight Benders Cuts

Convex-hull-based relaxation:

$$\begin{aligned} \min \quad & c_m^\top x_m + g_m^\top y_m + \theta_m \\ \text{s.t.} \quad & (z_m, x_m, y_m, \theta_m) \in \operatorname{conv}\left(\mathfrak{X}_m^i \wedge \{z_m \in \mathcal{Z}_n\}\right) \\ & z_m = x_n^i. \end{aligned} \tag{NA}$$

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- It is a linear program
- $\operatorname{conv}\left(\mathfrak{X}_m^i \wedge \{z_m \in \mathcal{Z}_n\}\right) \perp\!\!\!\perp x_n^i$
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$\rightarrow$ Valid & Tight Benders Cuts:

$$\theta_n \geq \overline{v}_n^i + (\overline{\lambda}_n^i)^\top x_n$$

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→ Valid & Tight Benders Cuts:

$$\theta_n \geq \overline{v}_n^i + (\overline{\lambda}_n^i)^\top x_n$$

- λ_m^i: optimal dual multiplier for (NA)
- $v_m^i := Q_m(x_n^i; \mathfrak{Q}_m^i) - (\lambda_m^i)^\top x_n^i$

$$\overline{v}_n^i := \sum_{m \in \mathcal{C}(n)} q_{nm} v_m^i;$$

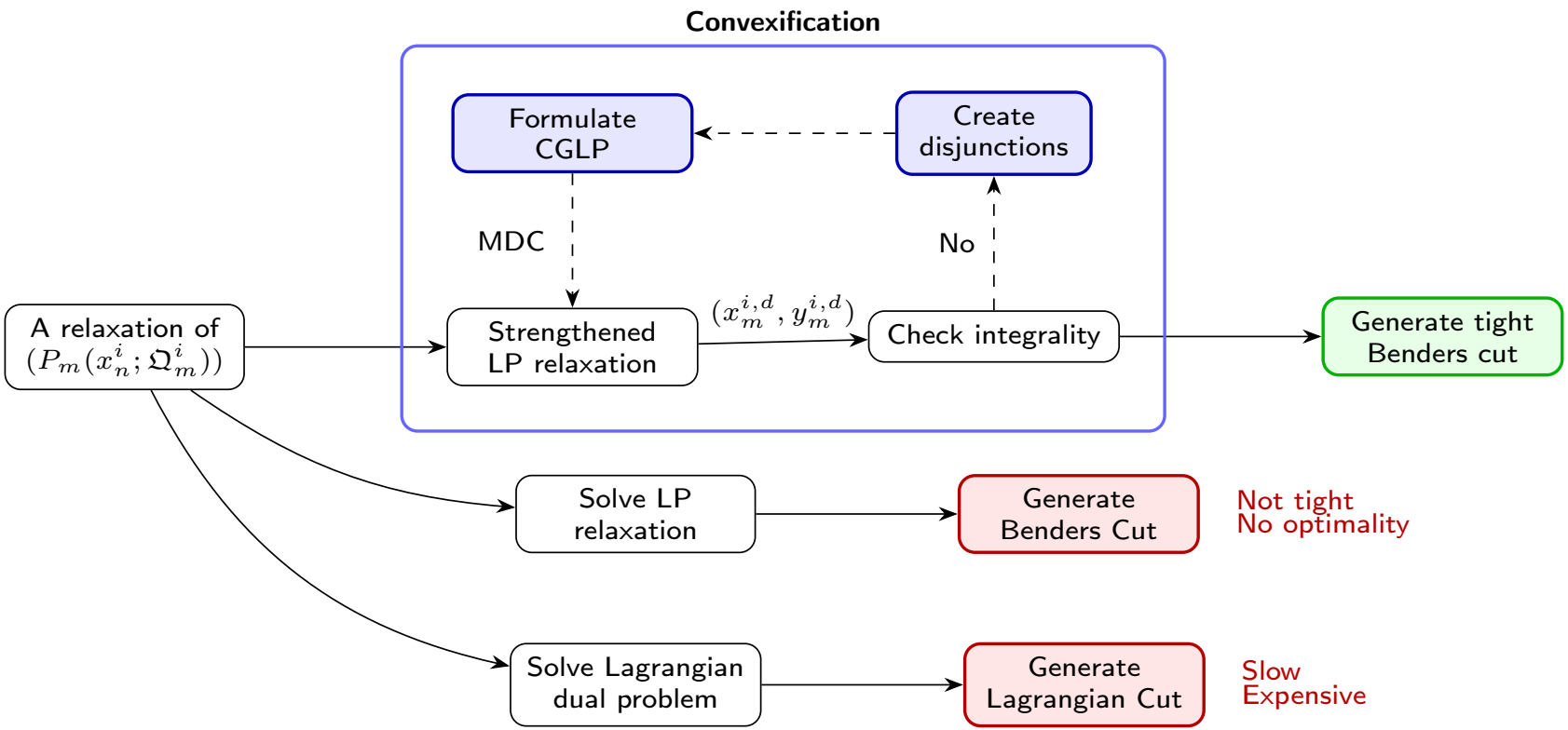
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Convexification

Approximating conv $\left(\mathfrak{X}_m^i \wedge \{z_m \in \mathcal{Z}_n\}\right)$

- LP Relaxation of $\mathfrak{X}_m^i \wedge \{z_m \in \mathcal{Z}_n\} \rightsquigarrow \text{LP}(\mathfrak{X}_m^i \wedge \{z_m \in \mathcal{Z}_n\})$;
- Utilize Branch-&-Bound tree to manage disjunctions;

$\rightsquigarrow$ Cut-generation linear program (CGLP) $\Rightarrow$ multi-term disjunctive cut (MDC).



Methodology: Cutting-Plane Tree (CPT) Algorithm

Given x_n^i, in the d-th iteration of the Branch-and-Bound (B&B) process:

a. Obtain $(x_n^i, x_m^{i,d}, y_m^{i,d}, \theta_m^{i,d})$ by solving LP relaxation of subproblem;

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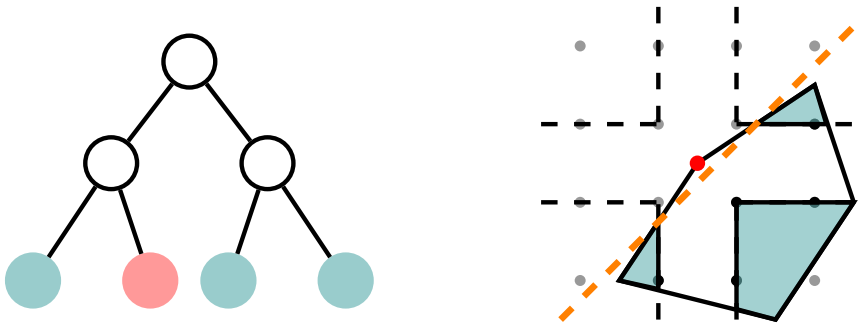


Figure: Green nodes are used to form a set of leaf nodes $\mathcal{L}_m^{i,d}$.

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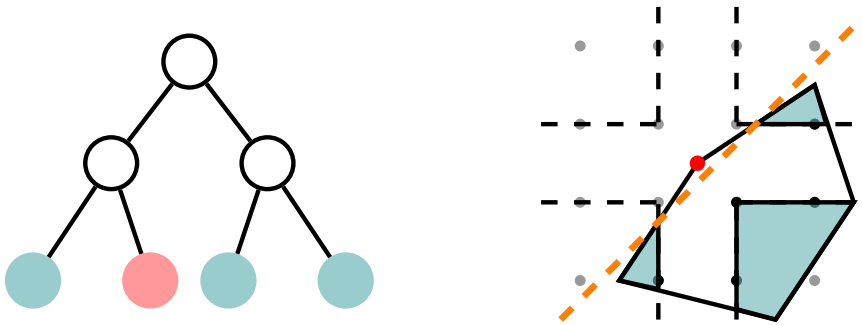


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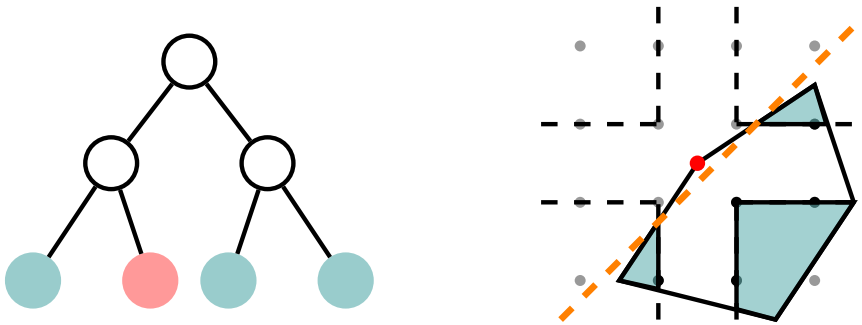


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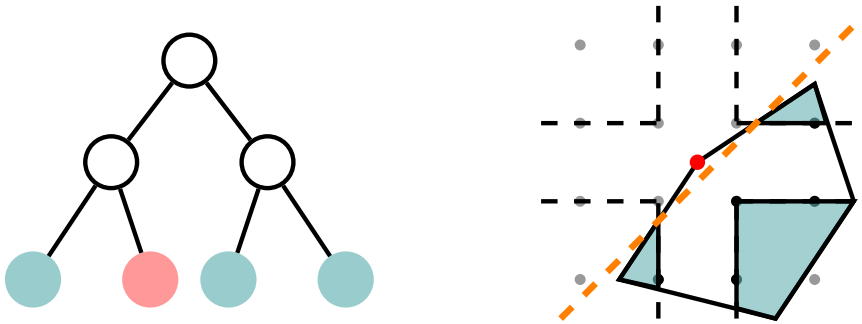


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These procedures will terminate within a finite number of iterations.

Multi-term Disjunctive Cuts (MDC)

The LP relaxation of the constraint system is of the form

$$\mathrm{LP}(\mathfrak{X}_m^i \wedge \{z_m \in \mathcal{Z}_n\}) = \left\{(z_m, x_m, y_m, \theta_m) \;\middle|\; B_m^i z_m + A_m^i x_m + C_m^i y_m + D_m^i \theta_m \geq b_m^i\right\}.$$

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A CGLP for generating MDCs given the leaf nodes $\mathcal{L}_m^{i,d}$ is formulated as:

$$\begin{aligned}
\max_{\substack{\pi_m, \pi_{m,0} \\ (\rho_m, \mu_m, \nu_m) \geq 0}} \quad & \pi_{m,0} - ({x_n^i}^\top, {x_m^{i,d}}^\top, {y_m^{i,d}}^\top, {\theta_m^{i,d}}^\top)\pi_m & \\
\text{s.t.} \quad & \pi_m = \left(B_m^{i,d}, A_m^{i,d}, C_m^{i,d}, D_m^{i,d}\right)^\top \rho_{m\tau} + \mu_{m\tau} - \nu_{m\tau} & \forall \tau \in \mathcal{L}_m^{i,d} \\
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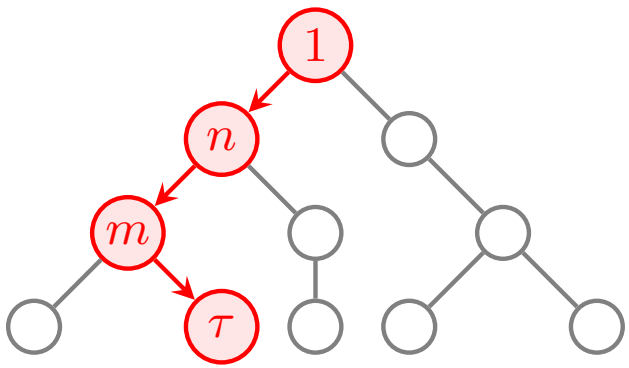
Multi-term Disjunctive Cut

$$(z_m^\top, x_m^\top, y_m^\top, \theta_m)\ \pi_m^* \ \geq\ \pi_{m,0}^*$$

Forward Pass

Forward Pass of Iteration i:

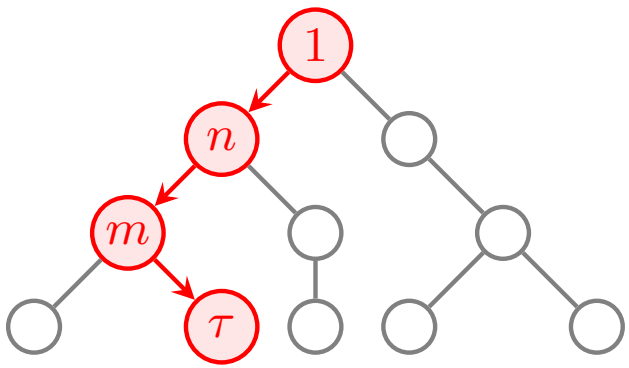
- Sample a scenario path $\mathcal{P}^{i,\omega} = \{1, n, m, \tau\}$;



Forward Pass

Forward Pass of Iteration i:

- Sample a scenario path $\mathcal{P}^{i,\omega} = \{1, n, m, \tau\}$;



- **For** each node $n \in \mathcal{P}^{i,\omega}$:

  Solve problem $(P_n(x^i_{a(n)}; \mathfrak{Q}^{i-1}_n))$:

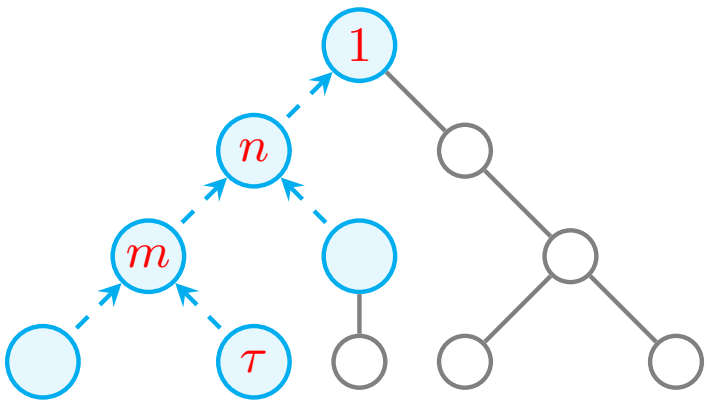
$$
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& z_n = x^i_{a(n)}.
\end{aligned}
$$

Backward Pass

Backward Pass of Iteration i:

- **For** each node $n \in \mathcal{P}^{i,\omega}$:
  - **For** each child node $m \in \mathcal{C}(n)$:

    Implement CPT algorithm to tighten m's LP relaxation;
  - Generate a tight Benders cut to refine $\mathfrak{Q}_n^{i-1}$ to $\mathfrak{Q}_n^i$;



Theorem

The algorithm defines an optimal solution to model (P) *within a finite number of iterations.*

Numerical Experiments

Generation Expansion Problem

Consider a multi-stage generation expansion problem (GEP):

$$
\begin{aligned}
\min \quad & \sum_{t=1}^{T}(c_t^1 x_t + c_t^2 y_t + p s_t) && && \text{(Investment cost + Generation cost)} \\
\text{s.t.} \quad & \sum_{s=1}^{t} x_s \leq \bar{u}, && \forall t, && \text{(Limitation on \# generators)} \\
& \mathbf{1}^\top y_t + s_t \geq d_t, && \forall t, && \text{(Demand satisfaction)} \\
& h_t N(\sum_{s=1}^{t} x_s + S_0) - y_t \geq 0, && \forall t, && \text{(Generation capacity)} \\
& x_t \in \mathbb{Z}_b^+, y_t \in \mathbb{R}_b^+, && \forall t.
\end{aligned}
$$

State x_t: Cumulative number of generators expanded;

Local y_t: Amount of generation; s_t: Amount of shortage;

- d_t: Random variable with R realizations.

Impact of Number of MDCs

D	T	R	LB	Gap(%)	Iter.	Time/Iter.	LB-time	LB-iter.
1	6	6	14804.5	40.8	198	3.1 (0.5)	7.9	3
		9	14755.7	40.0	243	2.5 (0.4)	4.6	3
	8	6	18443.8	13.1	177	3.4 (0.3)	11.6	4
		9	15472.8	42.9	177	3.4 (0.5)	7.0	3
2	6	6	14864.8	41.2	175	3.5 (1.1)	5.1	3
		9	14858.2	40.9	196	3.1 (0.6)	5.2	3
	8	6	17011.3	27.9	146	4.2 (1.2)	5.6	2
		9	15484.6	43.3	149	4.1 (0.7)	4.7	2
3	6	6	14783.5	36.4	180	3.4 (1.3)	5.5	3
		9	14702.0	36.8	254	2.4 (0.6)	5.6	3
	8	6	16889.1	32.1	138	4.4 (0.7)	15.0	5
		9	14703.7	39.9	170	3.6 (0.8)	3.9	2
i	6	6	18214.4	0.0	319	1.9 (0.6)	19.4	8
		9	18217.7	0.0	383	1.6 (0.2)	10.9	8
	8	6	19169.3	0.7	240	2.5 (0.3)	16.8	6
		9	19291.6	0.8	232	2.6 (0.3)	19.0	8

D: Maximum number of MDCs added to strengthen LP relax. in each iteration.

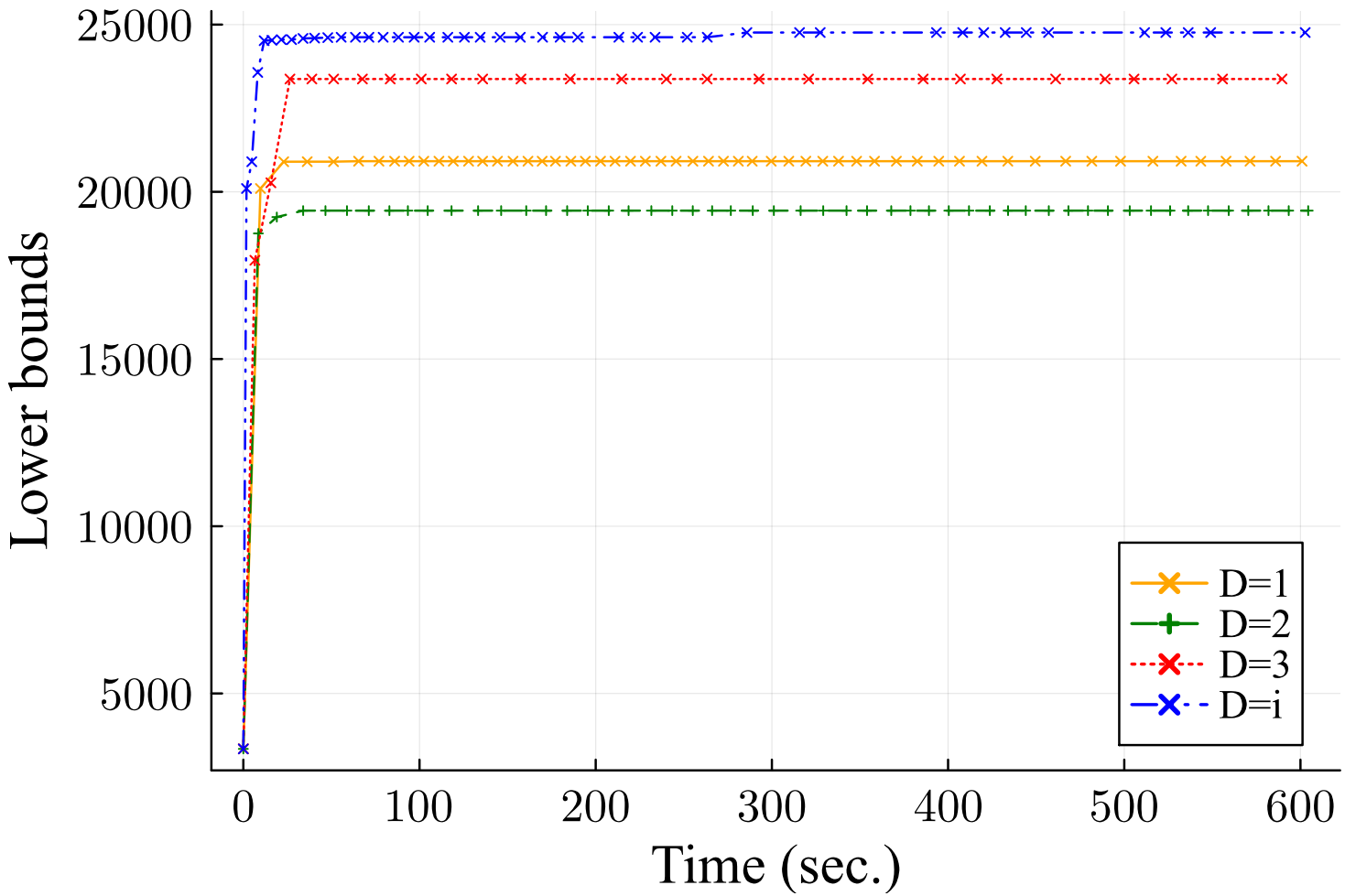
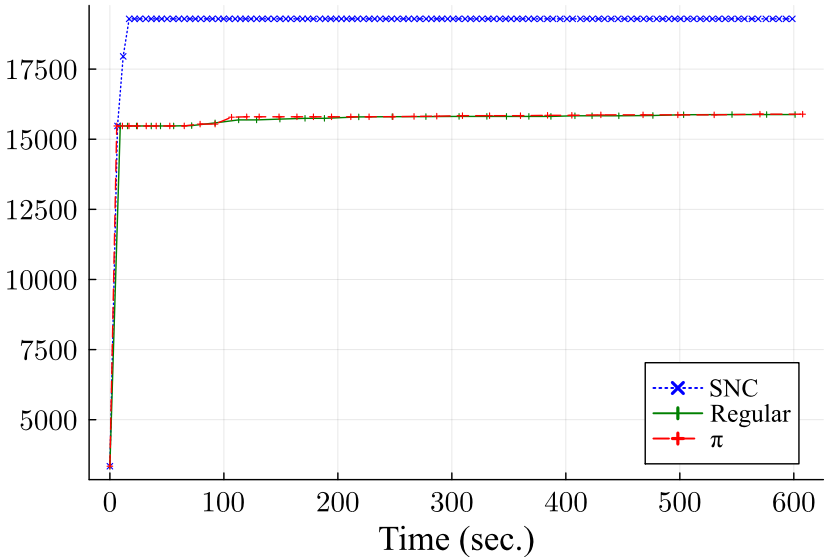
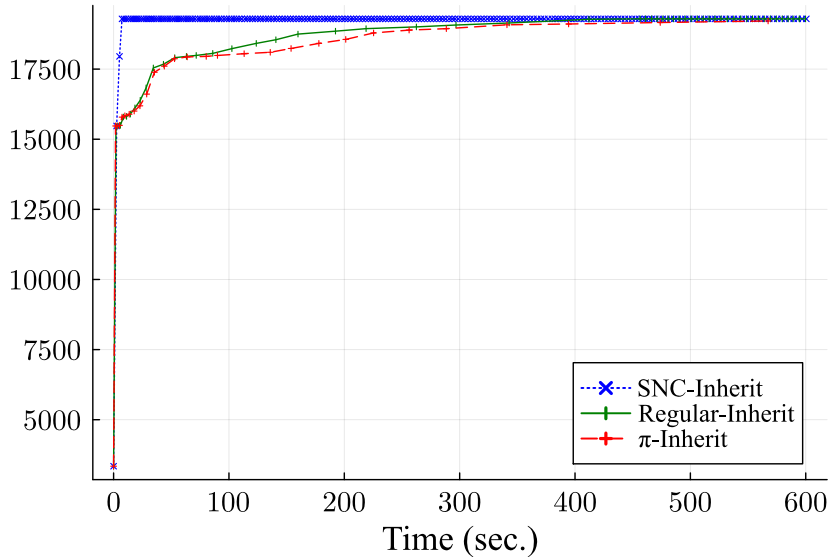


Figure: Evolution of the lower bounds using different numbers of MDCs.

Effect of Inheriting MDCs



(a) Without Inheritance.



(b) Inheriting MDCs.

Comparative Analysis of Cut Families

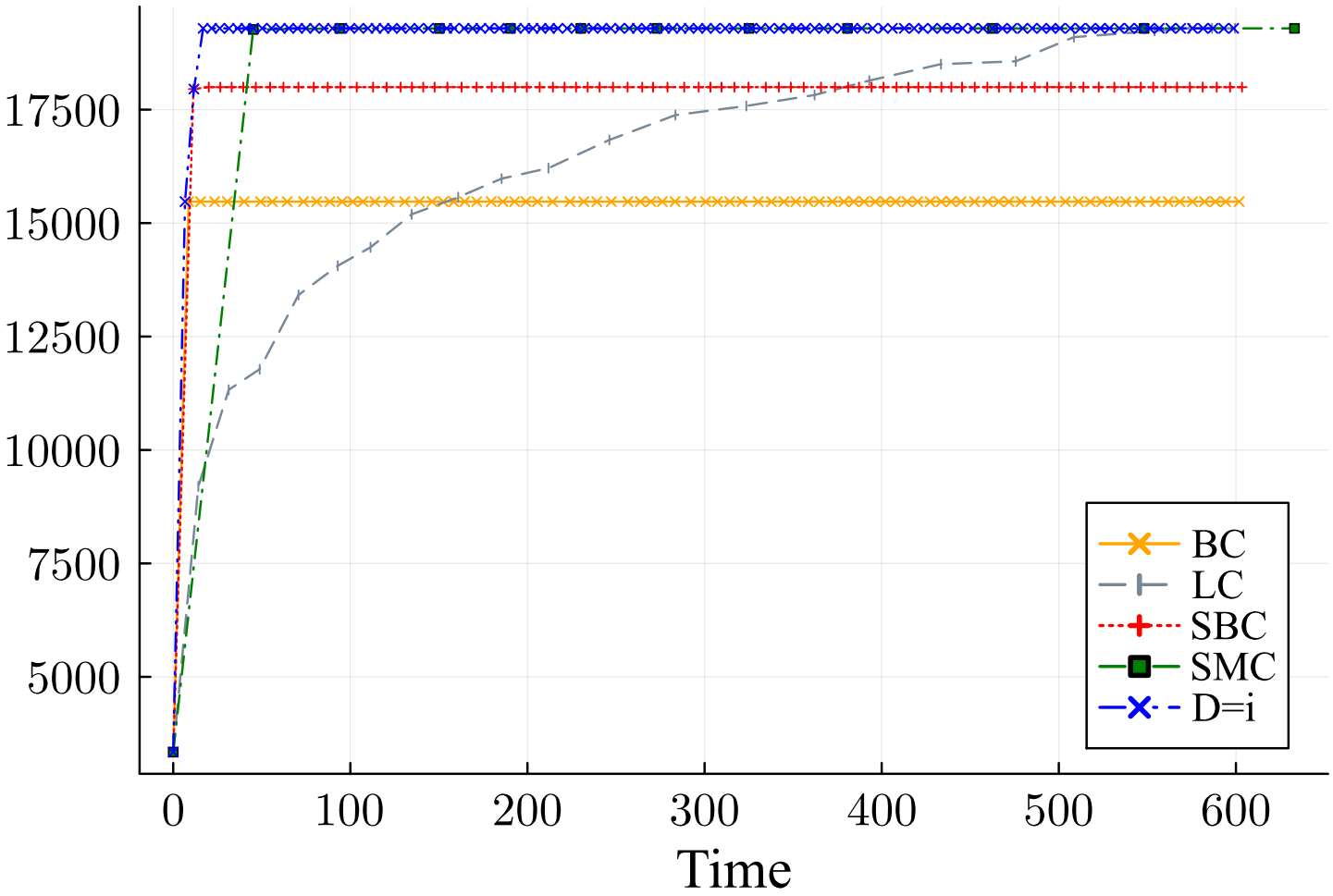


Figure: Evolution of the lower bounds for the GEP instance with $T = 8$ and $R = 9$ when using different cut families.

MSUC with RTS-GMLC (73 buses)

Cut	T	R	LB	Gap(%)	Iter.	Time/Iter.	LB-time	LB-iter.	#d
LC	6	5	51274.5	78.9	17	225.4 (122.0)	3140.4	16	-
		10	50636.8	78.8	12	348.0 (178.7)	3221.4	11	-
	8	5	52734.3	84.2	16	262.7 (140.3)	3941.1	16	-
		10	51957.3	83.8	12	346.0 (192.4)	3806.5	12	-
SMC	6	5	120566.4	52.0	6	776.6 (710.0)	3882.9	6	-
		10	104348.5	57.5	5	2309.1 (2477.5)	9236.3	5	
	8	5	160918.0	53.2	6	942.0 (990.3)	4709.9	6	-
		10	145323.4	55.9	5	2173.7 (2883.1)	8694.9	5	
π Inherit	6	5	237634.6	**0.9**	26	157.8 (153.8)	278.4	11	6.0 (3.3)
		10	233338.2	**1.1**	21	206.0 (196.5)	432.2	10	4.9 (2.7)
	8	5	297432.4	**9.2**	21	192.1 (195.4)	3841.5	21	4.8 (2.6)
		10	281197.2	11.7	17	246.2 (237.2)	3938.9	17	4.0 (2.2)
π ~~Inherit~~	6	5	231727.4	3.7	41	91.1 (62.6)	2065.1	32	4.6 (2.4)
		10	216267.9	8.6	34	115.0 (92.2)	3502.2	33	3.9 (2.0)
	8	5	294584.3	10.0	36	109.8 (84.2)	3844.4	36	4.1 (2.2)
		10	281304.5	**11.4**	28	133.9 (111.3)	3615.7	28	3.3 (1.7)

Outline

1 Introduction

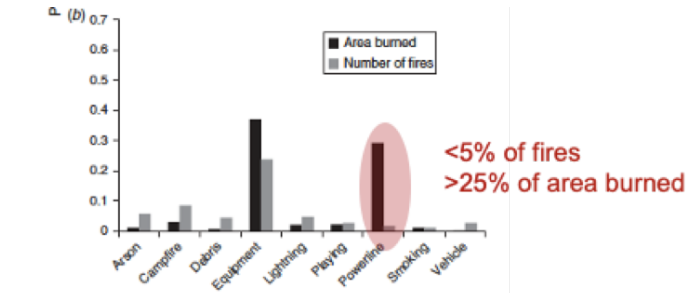
2 Lift-and-Cut Algorithm

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Wildfire Preventive Shutoff



Wildfires in San Diego County. Figure from Syphard, Alexandra D., and Jon E. Keeley. "Location, timing and extent of wildfire vary by cause of ignition." *International Journal of Wildland Fire* 24.1 (2015): 37-47.

(a) Powerline Damage



(b) Wildfire Spread

- Measures to prevent fire damages: De-energize risky components;
- Power shut-offs have significant impacts on society:
  - Outages at hospitals or chemical plants can lead to major accidents;
- Need to balance fire risk and outage loss:
  - Shutting off too early causes large economic and social costs;
  - Shutting off too late causes massive damage from wildfire spread.

Two-stage Model for Mitigating Wildfire Risk

Consider a multi-period power flow problem incorporating line de-energization decisions:

- Stochastic Disruptions:

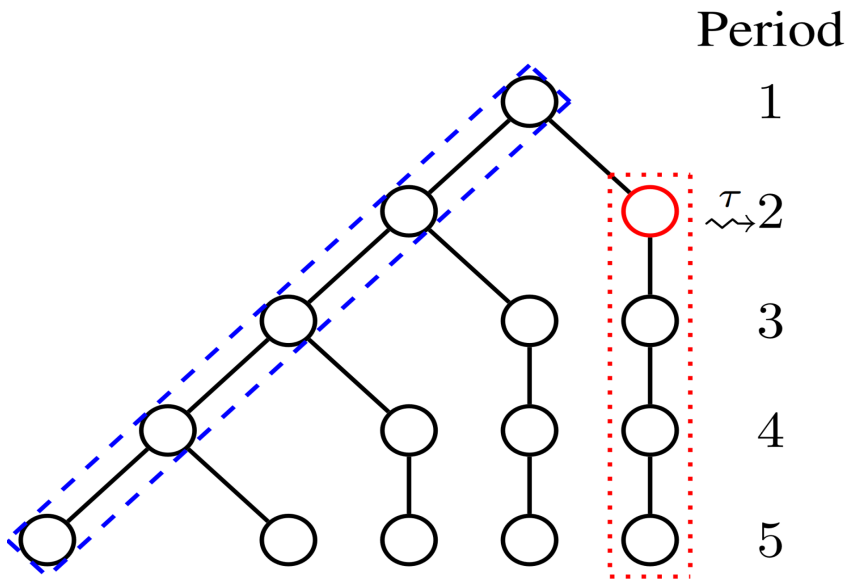
⇝ Timing, location, and magnitude.

- At most one Disruption:

⇝ Modeled as a two-stage stochastic mixed-integer program:

i) First-tage Nominal Plan before a wildfire occurs;

ii) Second-stage Disruption Plan after a disruption occurs.



The second-stage subproblem is:

$$\min \quad \sum_{t=\tau^\omega}^{T} \sum_{d\in\mathcal{D}} w_d(1-x_{dt}^\omega) + \sum_{c\in\mathcal{C}} r_c \nu_c^\omega$$

s.t. $\forall t \in \{\tau^\omega, \ldots, T\}$:

$$P_{ijt}^{L,\omega} \le -b_{ij}\left(\theta_{it}^\omega - \theta_{jt}^\omega + \bar{\theta}(1-y_{ij}^\omega)\right) \qquad \forall (i,j)\in\mathcal{L} \quad (2a)$$

$$P_{ijt}^{L,\omega} \ge -b_{ij}\left(\theta_{it}^\omega - \theta_{jt}^\omega + \underline{\theta}(1-y_{ij}^\omega)\right) \qquad \forall (i,j)\in\mathcal{L} \quad (2b)$$

$$-W_{ij}y_{ij}^\omega \le P_{ijt}^{L,\omega} \le W_{ij}y_{ij}^\omega \qquad \forall (i,j)\in\mathcal{L} \quad (2c)$$

$$\sum_{g\in\mathcal{G}_i} P_{gt}^{G,\omega} + \sum_{(i,j)\in\mathcal{L}_i} P_{ijt}^{L,\omega} = \sum_{d\in\mathcal{D}_i} D_{dt}x_{dt}^\omega \qquad \forall i\in\mathcal{B} \quad (2d)$$

$$\underline{P}_g^{G,\omega} y_g^\omega \le P_{gt}^{G,\omega} \le \bar{P}_g^{G,\omega} y_g^\omega \qquad \forall g\in\mathcal{G} \quad (2e)$$

$$y_i^\omega \ge x_{dt}^\omega \qquad \forall i\in\mathcal{B}, d\in\mathcal{D}_i \quad (2f)$$

$$y_i^\omega \ge y_g^\omega \qquad \forall i\in\mathcal{B}, g\in\mathcal{G}_i \quad (2g)$$

$$y_i^\omega \ge y_{ij}^\omega \qquad \forall i\in\mathcal{B}, (i,j)\in\mathcal{L} \quad (2h)$$

$$y_i^\omega \ge y_{ji}^\omega \qquad \forall i\in\mathcal{B}, (j,i)\in\mathcal{L} \quad (2i)$$

$$y_c^\omega \le z_c^\omega \qquad \forall c\in\mathcal{C} \quad (2j)$$

$$y_c^\omega \le 1-\nu_c^\omega \qquad \forall c\in\mathcal{C} \quad (2k)$$

$$\nu_c^\omega \ge v_c^\omega \qquad \forall c\in\mathcal{C} \quad (2l)$$

$$\nu_k^\omega \ge u_c^\omega z_c^\omega \qquad \forall c\in\mathcal{C}, k\in I_c^\omega \quad (2m)$$

$$z_c^\omega = \hat{z}_{c\tau^\omega-1} \qquad \forall c\in\mathcal{C} \quad (2n)$$

$$y_c^\omega, \nu_c^\omega, z_c^\omega \in \{0,1\} \qquad \forall c\in\mathcal{C}. \quad (2o)$$

Objective function:

- Minimize load-shedding cost;
- Minimize damage cost.

Power systems operations with shut-offs:

- Linearized Power Flow;
- Power Balance;
- Generator limits.

Logic Constraints:

Capture wildfire effects:

- How components are damaged or shut off;
- How fire spreads;
- How these affect system status.

Test System: RTS-GMLC

73-bus test system with geographic coordinates:

- Three levels of load priority and damage cost to generations;
- Visualization of "scenarios".

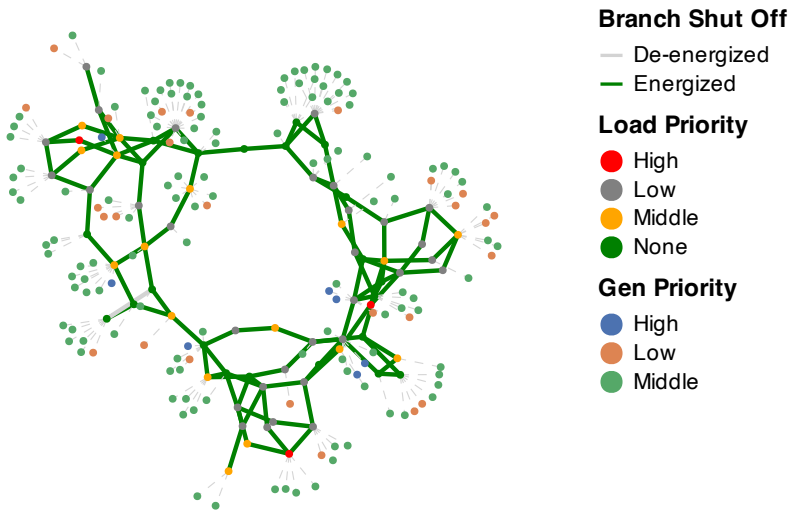


Figure: Power System Configuration.

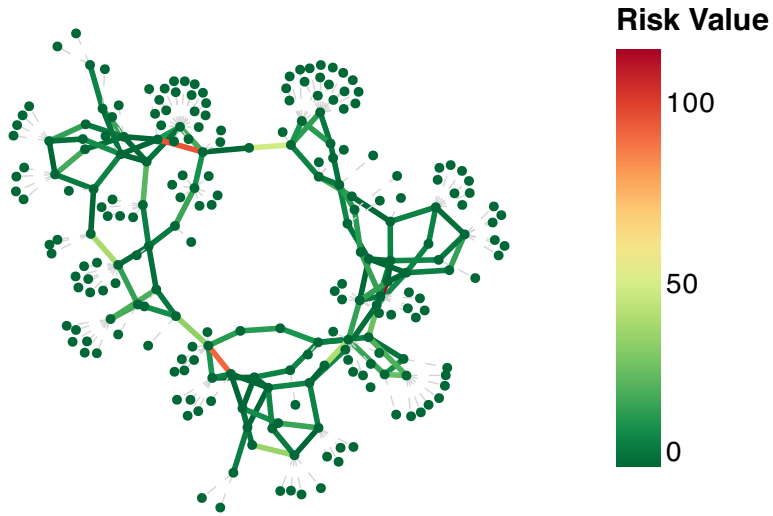


Figure: Wildfire risk value at $t = 13$.

Decomposition Algorithm with SMC

A sample size of 500 allows for sufficient convergence of upper and lower bounds:

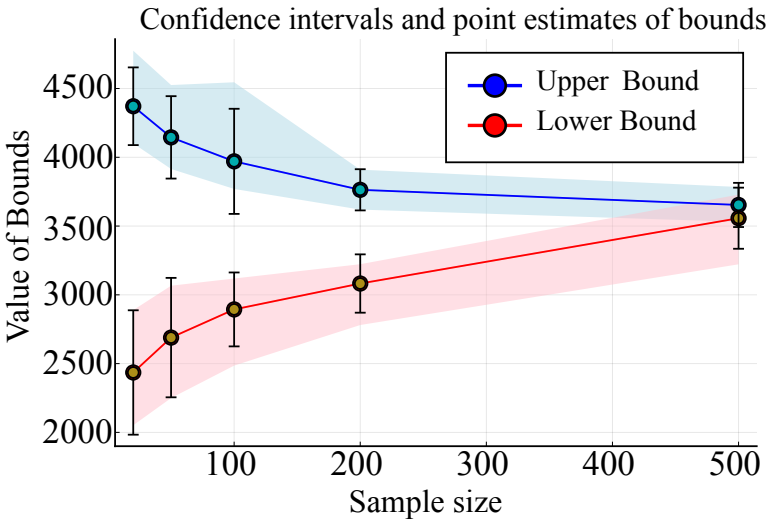


Table: Run-time (sec.) of utilizing SMC and LC with a tolerance of 1.0%, compared with a run-time of solving `extensive formulation` by Gurobi with MIPGap $= 1\%$.

Sample size	Algorithm Time		Gurobi Time
	SMC (Sec.)	LC (Sec.)	(Sec.)
20	712	3287	84
50	1248	8741	376
100	2604	17911	1510
200	5323	>18000	9186
500	12776	>18000	>18000

This shows:

- Convergence;
- Computational efficiency of the enhanced Lagrangian cuts (SMC).

Outline

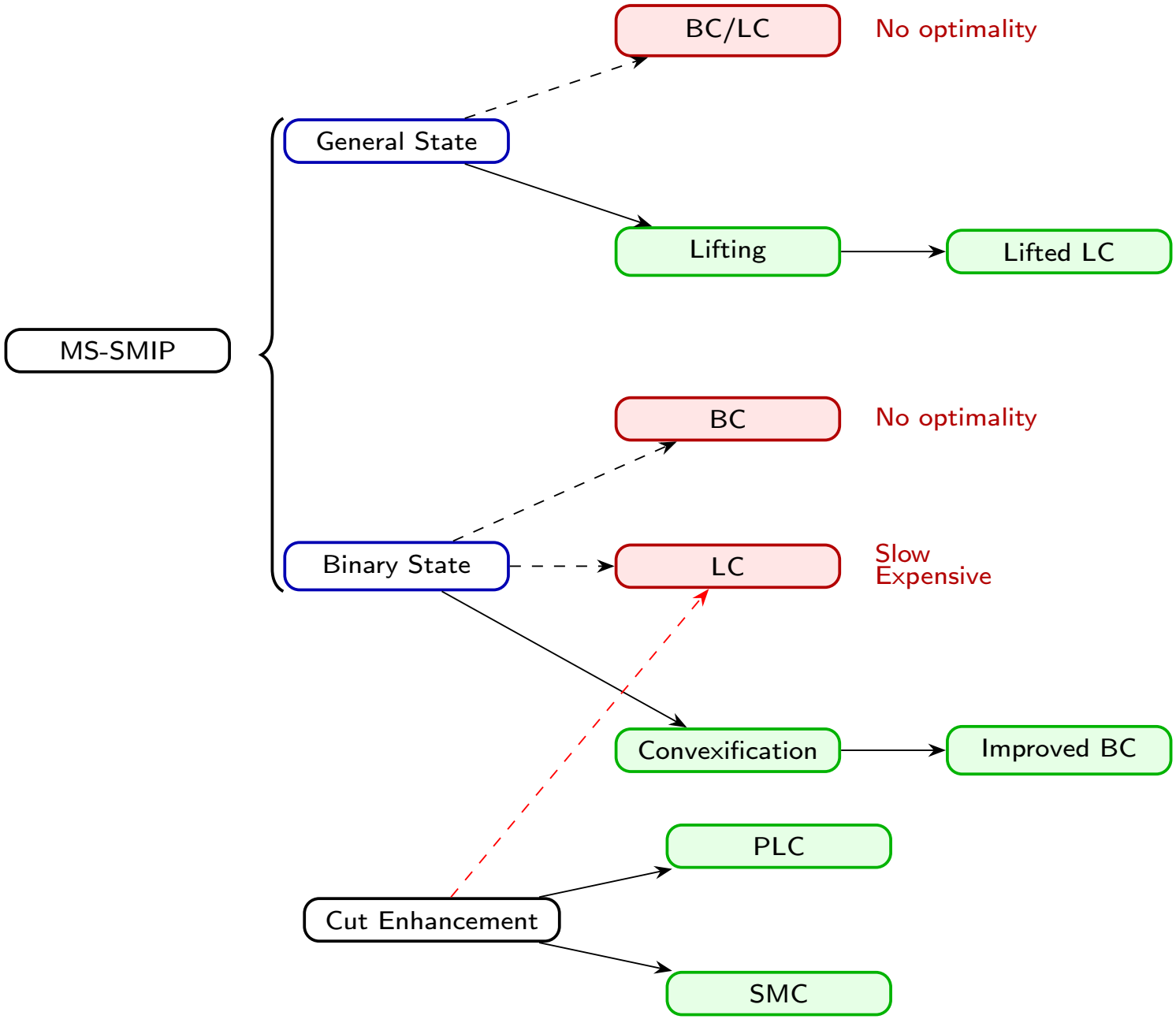
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5 Summary


MS-SMIP
General State
BC/LC
No optimality
Lifting
Lifted LC
Binary State
BC
No optimality
LC
Slow
Expensive
Convexification
Improved BC
Cut Enhancement
PLC
SMC

Summary

- **Lift-and-Cut Algorithm for MS-SMIP:**
  - Tightness of lifted Lagrangian cuts;
  - Guarantees global convergence (SDDP-L).
- **Enhanced Lagrangian Cuts:**
  - Pareto-optimality of PLC;
  - Heuristic method for generating SMC;
  - Significant numerical improvements.
- **Disjunctive Programming for Stronger Benders Cuts:**
  - Tightness of improved Benders cuts;
  - All cut-generation subproblems are linear programs;
  - Finite convergence for MS-SMIP with binary state variables.
- **Application:**
  - Model wildfires using stochastic disruption;
  - Stochastic programming models for wildfire risk mitigation;
  - Decomposition algorithms to solve large-scale instances.

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Proof of Global Convergence

Lemma

In every iteration $i \in \mathbb{N}$, the projection of lifted under-approximation $\phi_n^{L,i}(x_{a(n)}, \lambda_{a(n)})$ to the original state-variable space characterizes the piecewise convex envelope, i.e., for every iteration i,

$$\phi_n^i(x_{a(n)}^i; \mathbb{P}_n^i) = \mathrm{co}_{\hat{\mathcal{Z}}_{a(n)}^i}(\underline{Q}_n(\cdot; \Phi_n^i))(x_{a(n)}^i) \;\; \forall (x_{a(n)}^i, \lambda_{a(n)}^i) \in \mathfrak{Z}_n^i(\mathcal{Z}_{a(n)}, \mathbb{P}_{a(n)}^i),$$

where $\hat{\mathcal{Z}}_{a(n)}^i$ is the active region corresponding to $(x_{a(n)}^i, \lambda_{a(n)}^i)$.

Lemma

Suppose $\phi_m^i \xrightarrow{e} Q_m$ and $\phi_m^i \xrightarrow{p} Q_m$ for any $n \in \mathcal{N}, m \in \mathcal{C}(n)$. We have that $\underline{Q}_n^i(\cdot; \Phi_n^i)$ pointwise converges to Q_n.

Proposition

For every iteration $i \in \mathbb{N}$ at every node $n \in \mathcal{N}$, suppose $\underline{Q}_n(\cdot; \Phi_n^i) \xrightarrow{p} Q_n(\cdot)$. The sequence $\{\underline{Q}_n(x_{a(n)}; \Phi_n^i)\}$ is asymptotically equi-lower semi-continuous (equi-lsc) everywhere over $x_{a(n)} \in \mathcal{Z}_{a(n)}$.

Proof by contradiction.

Proposition

For node $n \in \mathcal{N}/\{1\}$, $\phi_n^i \xrightarrow{p} Q_n$ and $\phi_n^i \xrightarrow{e} Q_n$.

Proof by backward induction and discussion case by case. First, establish the pointwise convergence. Then establish asymptotically equi-lsc, which implies the epi-convergence.

Theorem

The SDDP-L algorithm generates a sequence of optimal solutions $\{(x_1^i, y_1^i)\}_{i\in\mathbb{N}}$ *to the root node problem, and, with probability one, we have*

$$V^* = \lim_{i\to\infty} \left(c_1^\top x_1^i + g_1^\top y_1^i + \sum_{m\in\mathcal{C}(1)} \underline{Q}_m \left(x_1^i; \Phi_m^i\right) \right),$$

and every accumulation point of the sequence $\{(x_1^i, y_1^i)\}_{i\in\mathbb{N}}$, (x_1^*, y_1^*), *is an optimal solution.*

Proof.

Given $\phi_n^i \xrightarrow{p} Q_n$ and $\phi_n^i \xrightarrow{e} Q_n$ for all $n \in \mathcal{N}\backslash\{1\}$, Theorem 7.33 in [Rockafellar and Wets, 2009] holds for the first-stage problem because we have $\phi_m^i \xrightarrow{e} Q_m$ for all $m \in \mathcal{C}(1)$ and both ϕ_m^i and Q_m are bounded and lower semi-continuous, which leads to convergence in optimal solutions and lower bounds. □